Qualcomm Technologies, Inc.

QRB5165 Hardware Design Guide

Hardware Design Guide

80-PV086-5P Rev. AA

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1 QRB5165 overview

The Qualcomm Robotics RB5 Platform supports the development of high-compute, AI-enabled, low power robots and drones for the consumer, enterprise, defense, industrial, and professional service sectors that can be connected by 5G. The QRB5165 processor, customized for robotics applications, offers a powerful heterogeneous computing architecture to efficiently run complex AI and deep learning workloads and on-device edge inferencing while using lower power, on device machine learning, and accurate edge inferencing. The processor also offers a powerful image signal processor (ISP) with support for multiple concurrent cameras, a dedicated computer vision engine for enhanced video analytics (EVA), as well as the new Qualcomm® Hexagon™ Tensor Accelerator (HTA). With support for 4G and 5G connectivity speeds via a companion module, the Qualcomm Robotics RB5 platform helps pave the way for the proliferation of 5G in robotics and intelligent systems.

1.1 QRB5165 functional block diagram

The following figure shows the QRB5165 functional block diagram.

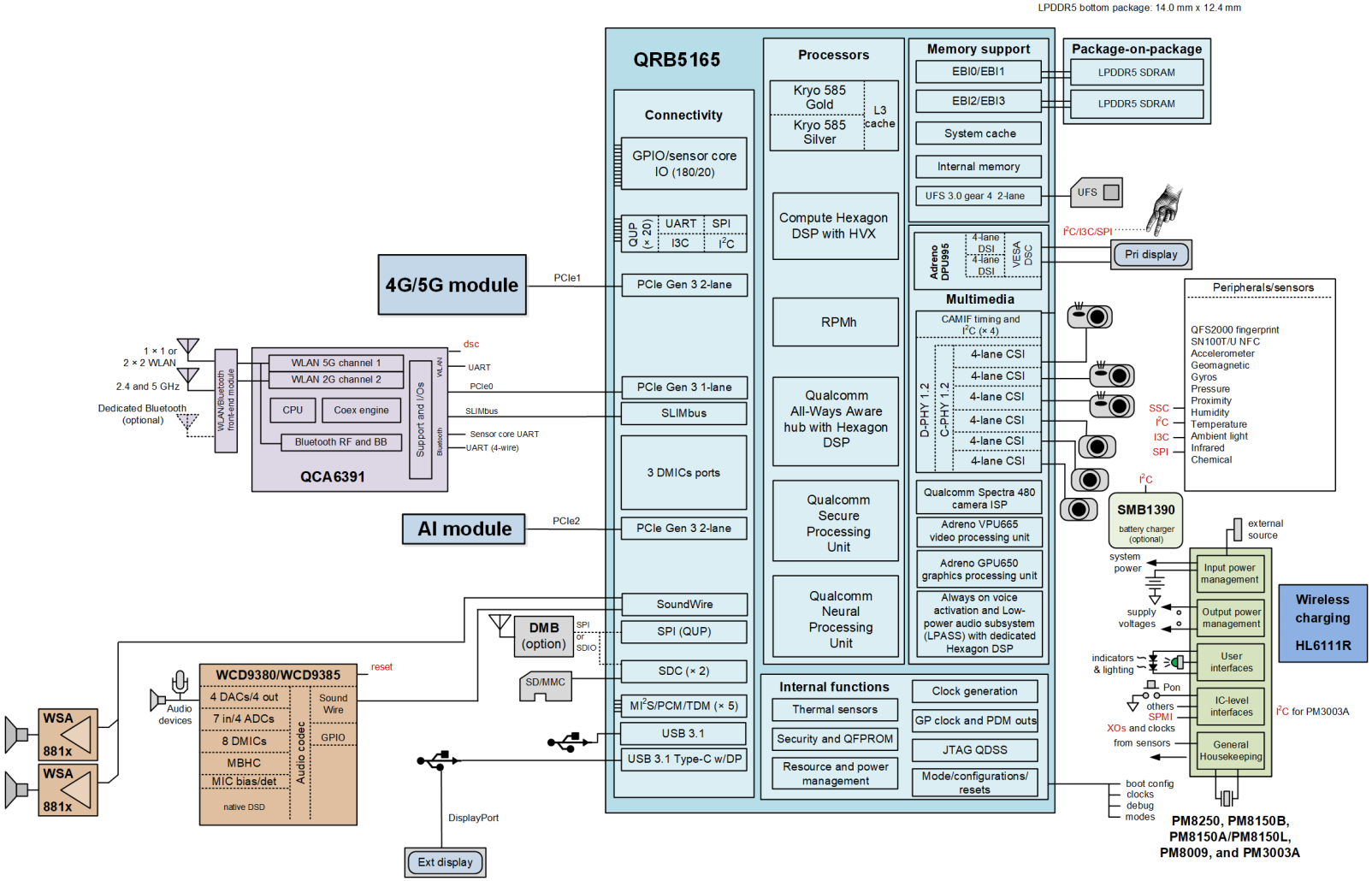


Figure 1-1 QRB5165 functional block diagram

1.2 QRB5165 subsystem

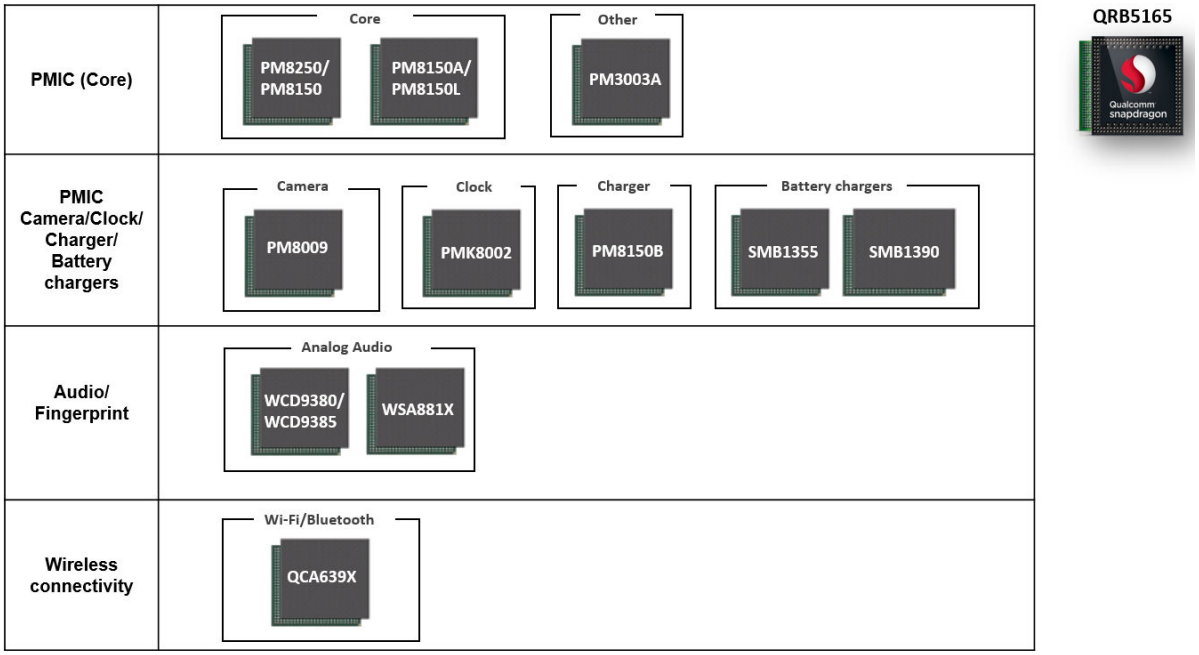
The QRB5165 + 4G/5G module consists of two subsystems:

- QRB5165 AP/Wi-Fi/Audio subsystems
- 4G/5G module

Table 1-1 QRB5165 subsystem

Processor	PMIC	Wi-Fi(11ax)/Bluetooth®	Audio
QRB5165	PM8250 PM8150B PM8150A or PM8150L PM3003A-5 PM8009-1 PMK8002 SMB1355 or SMB1390	QCA6391 (FCNSP) or QCA6390 (Bumped die)	WCD9380/WCD9385 WSA881x

NOTE SMB1355 or SMB1390 is optional for parallel charging with additional charging capability.



1.3 QRB5165 architecture overview

The following figure captures the major architecture blocks of QRB5165:

1. Kryo subsystem
2. Always-on subsystem (AOSS)
3. Multimedia subsystem (MMSS)

4. Low-power audio subsystem (LPASS)
5. Peripheral subsystem (PSS)
6. Qualcomm Sensing Hub
7. Memory support and the bus system that supports all major blocks
8. Compute subsystem
9. Qualcomm secure processing unit
10. Qualcomm neural processing unit

NOTE Details regarding these architectural blocks are interspersed with the hardware details in SoC subsystems.

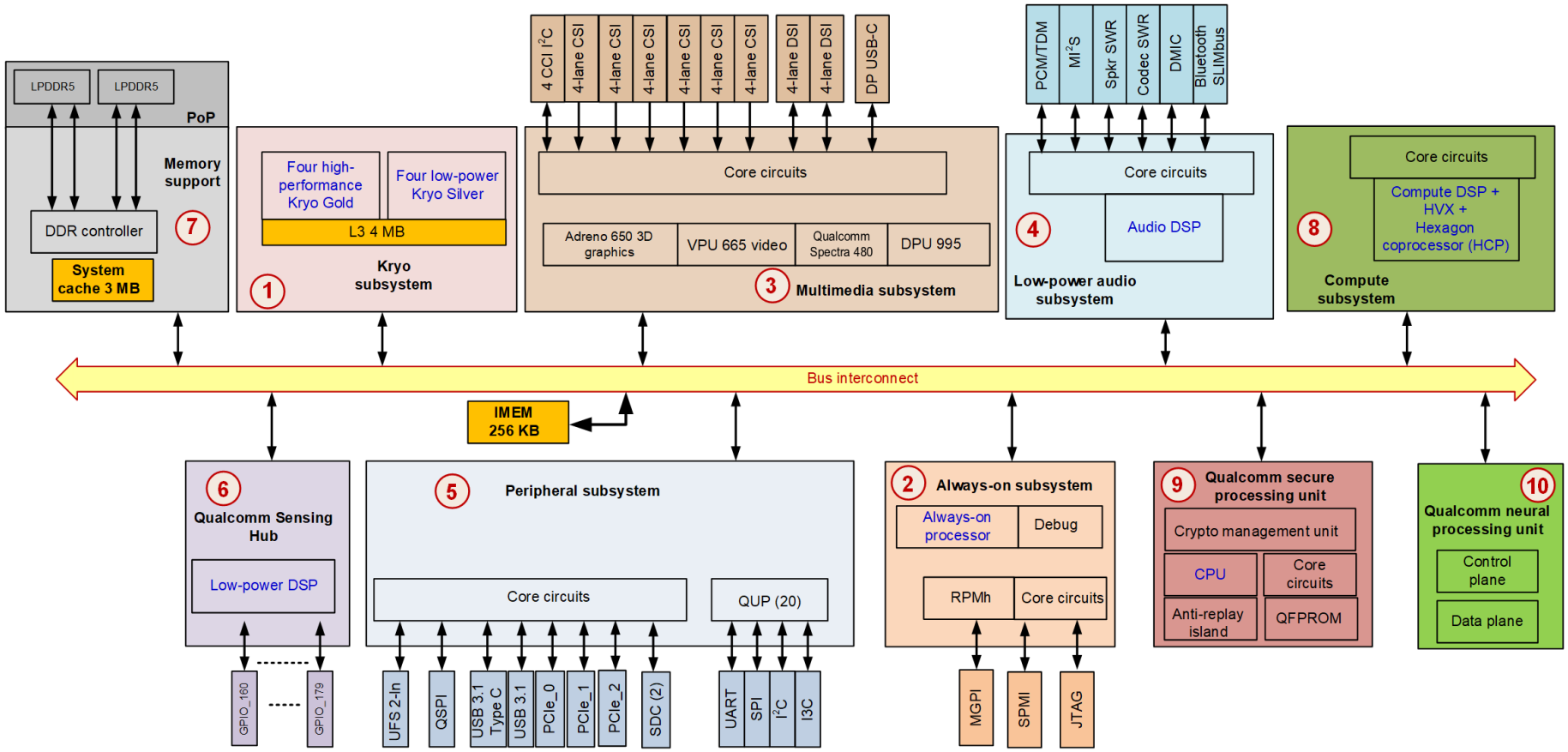


Figure 1-2 QRB5165 architecture overview

1.4 QRB5165 features

The following table lists all the features and QRB5165 capabilities.

NOTE Some of the hardware features integrated within the QRB5165 must be enabled by software. See the latest revision of the applicable software release notes to identify the enabled QRB5165 features.

Table 1-2 QRB5165 feature capabilities

Feature	QRB5165 capability
Processors	
Applications	Kryo 585 – 64-bit applications processor with a 4 MB L3 cache ■ Quad high-performance Kryo Gold cores □ Three Kryo Gold cores with a 256 KB L2 cache per core, Fmax at 2.419 GHz □ One Kryo Gold prime core with a 512 KB L2 cache, Fmax at 2.842 GHz. ■ Quad low-power Kryo Silver cores with a 128 KB L2 cache per core, Fmax at 1.805 GHz.
Digital signal processing	Compute Hexagon DSP with quad Hexagon Vector eXtensions (quad-HVX) and Hexagon Coprocessor (Hexagon CP) 2.0 ■ Used for video playback enhancements, virtual reality, computer vision, camera snapshot enhancements, video capture enhancement, machine learning, and so on. ■ The HCP is a vision and imaging hardware accelerator to offload and accelerate the Hexagon software algorithmic functions. Audio Hexagon DSP dedicated to an audio subsystem. Sensor Hexagon DSP in the Qualcomm®Sensing Hub to support always-on, low-power use cases. All Hexagon DSP are cache-based processors with full access to DDR memory for large memory requirements.
Always-on system	Always-on subsystem with always-on processor. Hardware-based resource and power management (RPMh) with hardware accelerators for voltage control and regulation, clock management, and resource communication.
Artificial intelligence	Qualcomm NPU230 dedicated neural processing unit for performance and always-on neural network (NN) use cases. It incorporates an NN matrix engine to ensure efficient execution of various neural networks and their parameters. The NPU may be used for typical imaging, video, audio, and data-based NN use cases and will typically be used in conjunction with the compute Hexagon DSP subsystem.
Memory support	
System memory via PoP and EBI	Four-channel PoP high-speed memory – LPDDR5 SDRAM (4 × 16-bit) designed for a 2750 MHz (LPDDR5) clock and system cache
External memory via UFS	UFS 3.1 Gear 4 – for on-board memory
External memory via SDC	SD v3.0 4-bit for SD card
Multimedia	

Table 1-2 QRB5165 feature capabilities (cont.)

Feature	QRB5165 capability
Adreno display processing unit (DPU)	Adreno DPU 995, supports up to three 4K display (one internal display through DSI and two external displays through DisplayPort)
Display interface	Two 4-lane DSI D-PHY 1.2 or two 3-trio C-PHY v1.1 with VESA DSC v1.1 DisplayPort v1.4 at 8.1 Gbps/lane over Type-C with support for MST and VESA DSC v1.1 and FEC (USB3 and USB2 concurrency supported) Miracast – up to 4K60
Display performance	5040 × 2160 at 60 Hz (or 120 Hz in VR mode), up to 30 bpp Two 2560 × 2560 at 120 Hz for dual-panel VR displays, up to 30 bpp 4096 buffer width, and 16 hardware-layer composition
Display processing	Qualcomm® TruPalette Display Feature – HDR10+ and HDR10 tone mapping, color gamut mapping, six-zone, memory color, and picture adjust
Pixel processing	Qualcomm® Low-Power Picture Enhancement display compression [Qualcomm Universal Bandwidth Compression (UBWC 4.0, DSC v1.1)], CABL, FOSS, Qualcomm Local Tone Mapping, QSync, and destination scaler with DE
Camera performance	Qualcomm Spectra 480 ISP to support up to 12 cameras by D-PHY and 18 cameras by C-PHY (seven concurrent) ■ Real-time sensor input resolution: 25 + 25 + 2 + 2 + 2 + 2 + 2 ■ 64 MP 30 fps ZSL with a dual ISP ■ Hardware 2PD support and improved face detection ■ 4K120 camcorder and improved spatial noise reduction
Camera interface	MIPI CSI configurable in 4 + 4 + 4 + 4 + 4 + 4 configuration ■ D-PHY v1.2: 2.5 Gbps/lane on four lanes per port ■ C-PHY v1.2: 10.26 Gbps/trio on three trios per port
Adreno video processing unit (VPU)	■ Adreno VPU 665 – fifth-generation UHD video processing unit ■ Video decode up to 4K240/8K60 ■ Video encode up to 4K120/8K30 ■ Concurrent 4K60 decode and 4K30 encode for wireless display ■ Native decode support for H.265 Main 10, H.265 Main, H.264 High, VP9 profile 2, VP8, and MPEG-2 codecs ■ Native encode support for H.265 Main 10, H.265 Main, H.264 High, and VP8 codecs ■ Improved encoder with up to 30% reduction in bit rate for same subjective quality video ■ New computer vision processor (CVP) for object detection and tracking
Adreno graphic processing unit (GPU)	■ Adreno GPU 650, Fmax at 587 MHz – 4K 60 fps UI or 2x 2K 60 fps UI ■ OpenGL ES 3.2, Vulkan 1.1, DX12 FL 12_1 ■ OpenCL 2.0 full profile
Audio codec	Integrated within the WCD9380/WCD9385 high fidelity audio codec: ■ Four DACs; four outputs ■ Seven differential analog inputs; four ADCs ■ Eight digital microphones ■ High dynamic range recording ■ HPH load equalization

Table 1-2 QRB5165 feature capabilities (cont.)

Feature	QRB5165 capability
	■ Native DSD, MBHC, and ANC ■ 122 dB dynamic range for HPH ports ■ 32-bit DAC ■ 44.1 kHz family native playback ■ Four GPIOs
Speaker amplifier	Integrated within the WSA8810/WSA8815 class D/G, low noise smart amplifier: ■ 2 W/4 W output power into 8 Ω load ■ Integrated SmartBoost ■ Integrated feedback speaker protection for excursion and temperature control of the transducers ■ Support for receiver assist speaker (RAS) or speaker as receiver (SAR) with handset ANC
Low-power audio subsystem (LPASS)	■ Essential voice communications package ■ Advanced voice communications package ■ Voice UI voice activation package ■ Voice UI speech enhancement package ■ 3D audio capture package
Audio interfaces	SLIMbus: ■ QCA639x SLIMbus SWR: ■ SoundWire interface (two Tx and two Rx data lines; optional configuration of three Tx and one Rx data line) for codec ■ Dedicated SoundWire interface for smart speaker amplifier Digital Mic: ■ Three DMIC ports support up to six DMICs MI^{2}S: ■ Five MI^{2}S with 2x data lanes to support full duplex stereo, or up to 4 channel Tx/Rx application ■ One MI^{2}S supports 4 data lanes for up to 8 channels Tx/Rx application TDM/PCM: ■ Up to 32 channels per individual interface ■ Short, long, and one-slot sync mode ■ Maximum clock frequency of 24.576 MHz
Digital mobile broadcast (DMB)	External IC required with SPI or SDIO interface
Connectivity	
Qualcomm universal peripheral (QUP) ports	20:7 bits each for four QUPs and 4-bit each for the other QUPs; multiplexed serial interface functions
UART	UART interface available on all QUPs. HS-UART available on GPIO QUP6/QUP7/ QUP12/QUP13/QUP18/QUP19/Qualcomm Sensing Hub
I^{2}C	I^{2}C interface available on all QUPs up to 1 Mbps for touch, sensors, near field communicator (NFC), and so on; dedicated controller for each port

Table 1-2 QRB5165 feature capabilities (cont.)

Feature	QRB5165 capability
I3C	I3C interface available on GPIO QUP0/QUP1/QUP8/QUP14 and Qualcomm Sensing Hub QUP0/QUP1/QUP2. I3C IBI (in-band interrupt) will not be supported.
SPI	SPI interfaces available on all QUPs for cameras, sensors, and so on; dedicated controller for each port
CCI I^2C	Four dedicated I^2C interfaces for camera
USB	Two USB 3.1 ports, support Type-C with DisplayPort v1.4 in one port
Secure digital interfaces	■ Two 4-bit ports (SDC2 and SDC4); SD 3.0 ■ SDC2 is dual-voltage ■ SD/MMC card and DMB
Wireless connectivity	QCA639x 802.11ax QCA643x/QCA642x 802.11ad at 60 GHz
Touchscreen support	Capacitive panels via ext IC (I^2C, I3C, SPI, and interrupts)
Fingerprint support	Ultrasonic Qualcomm® Fingerprint Sensors for under glass, under metal, or under OLED display. QFS2000/QFS2080/QFS2580/QFS2530 modules.
Configurable GPIOs	
Number of GPIO ports	180– GPIO_0 to GPIO_179
Input configurations	Pull-up, pull-down, keeper, or no pull
Output configurations	Programmable drive current
Top-level mode multiplexer	Provides a convenient way to program groups of GPIOs
Internal functions	
Security	
General hardware security features	SPU-240 with planned certification enabling Android Strongbox and iUICC, Secure Boot 3.0, Debug security, Key provisioning security, TrustZone, Qualcomm® Trusted Execution Environment v5, hardware-backed keystore, combined image signing, image encryption, secure peripherals, Inline crypto engine (ICE), file-based encryption (FBE)
Crypto engines	Crypto engine v5 (CE5), Qualcomm to submit for Federal Information Processing Standards (FIPS) certification to Certified for FIPS 140-2.
TrustZone services	Secure file system, Fast Trusted Storage
DRM support in hardware QFPROM	PlayReady SL2000/SL3000, Widevine level 1 and level 3, ISDB-T fuse bits available for OEM use
Access control	Programmable security domain protection and sandboxing
Boot sequence	1. Applications PBL 2. XBL 3. SHRM 4. AOP 5. HLOS 6. Rest of subsystems ■ Emergency boot over USB 3.1

Table 1-2 QRB5165 feature capabilities (cont.)

Feature	QRB5165 capability
PLLs and clocks	■ Multiple clock regimes; watchdog and sleep timers ■ Input: 19.2 MHz CXO ■ General-purpose outputs: M/N counter and PDM
Debug	JTAG, design for software debug (DFSD), embedded USB debug (EUD)
Others	Thermal sensors; modes and resets; peripheral subsystem
Chipset interface features	
Power management	2-line SPMI; plus other lines, as needed, via GPIOs, I^2C
Wireless connectivity	
WLAN Bluetooth	PCIe interface SLIMbus/UART interface
Fabrication technology and package	
Digital die	7 nm process
PoP – small, thermally efficient package	MPSP1099 (QRB5165-LPDDR5): 14.0 × 12.4 × 0.56 mm max (without memory device on top)
QRB5165-LPDDR5	
Bottom pin array Top pin array	Bottom: 1099-pin picoscale package (1099 PSP); 0.35 mm pitch Top: 496-pin nanoscale package (496 NSP); 0.4 mm pitch

[a] Higher resolution and wider aspect ratio displays can be supported. Exact panel details and timings are required to determine if it can be supported.

2 Clock distribution

A single 38.4 MHz crystal could be the source for the entire system. The clock drivers have a 50 Ω output impedance. The real-time clock is backed up with a coin cell or capacitor. More clocks are available at the GPIO ports than are used in this example.

DIV_CLK outputs are available at the SLEEP_CLK rate or 38.4 MHz XO rate divided by 2, 4, 8, or 16. Low-power baseband clocks have tight duty-cycle values near 50%. The clock outputs can be controlled by software (via SPMI).

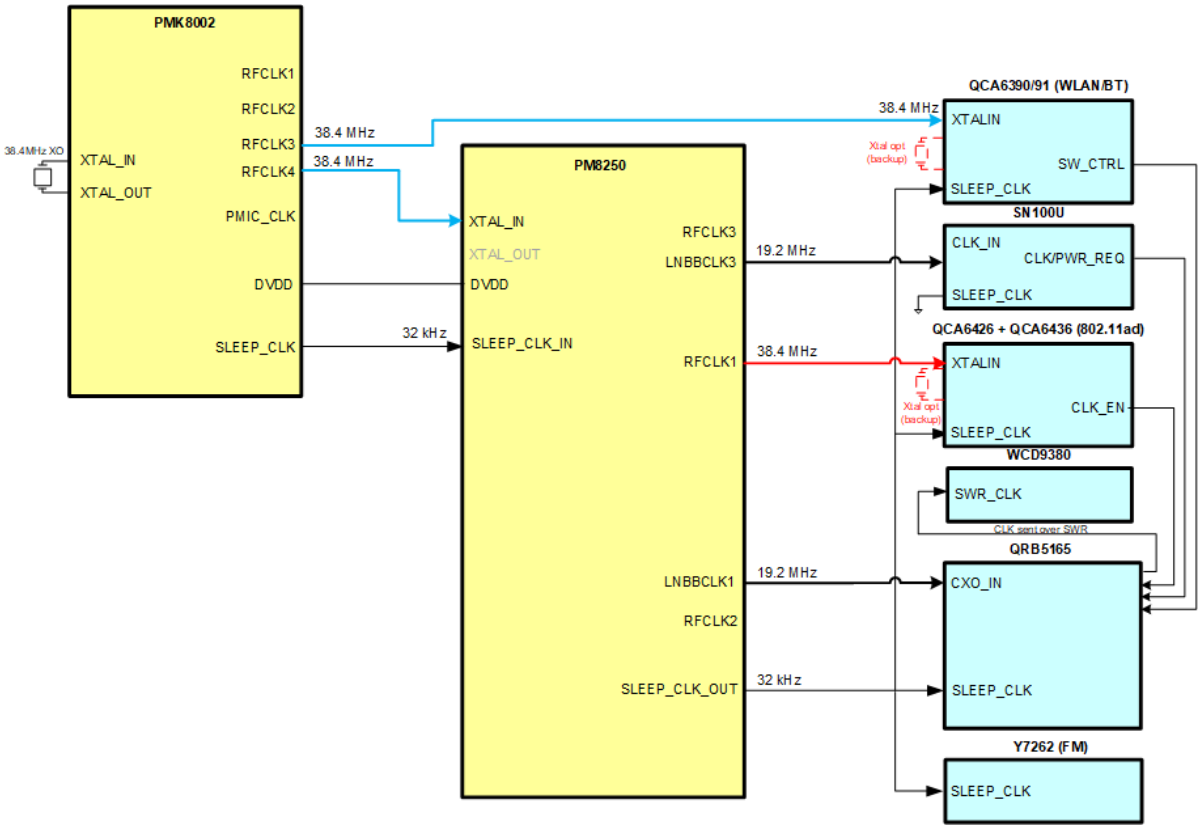


Figure 2-1 QRB5165 system-wide clock distribution

3 Boot configuration and memory

3.1 Boot flow

The following figure displays the boot flow sequence of events.

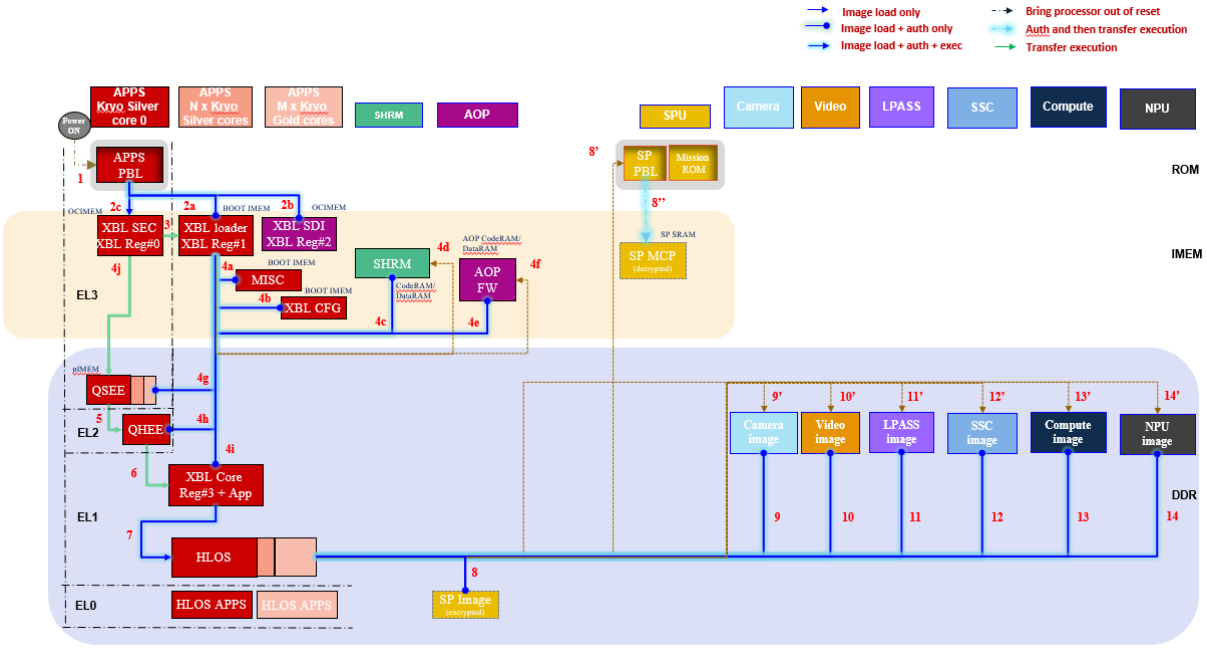


Figure 3-1 QRB5165 boot flow

Table 3-1 QRB5165 boot flow

Component	Based on processor	Loaded from	Executes in
APPS PBL (Application primary boot loader)	Kryo Silver core 0	N/A	ROM
XBL loader (eXtensible boot loader)	Kryo Silver core 0	UFS	BOOT IMEM
MISC (Multi image signature and integrity check) image	Kryo Silver core 0	UFS	BOOT IMEM, LPDDR5
Qualcomm® Trusted Execution Environment	Kryo Silver/Gold cores	UFS	OCIMEM, LPDDR5 (pIMEM)

Table 3-1 QRB5165 boot flow (cont.)

Component	Based on processor	Loaded from	Executes in
Hypervisor execution environment	Kryo Silver core 0	UFS	LPDDR5
SHRM		UFS	SHRM CodeRAM, System cache
AOP_FW, XBL-SDI	Always-on processor	UFS	AOP code RAM
XBL Core (UEFI) + Application	Kryo Silver core 0	UFS	LPDDR5
HLOS	Kryo Silver/Gold	UFS	LPDDR5
SP PBL (Secure processor primary boot loader)	SPU	–	ROM
SP mission ROM	SPU	–	ROM
SP image	SPU	UFS	SP SRAM
Camera, Video, LPASS, Snapdragon sensor core (Qualcomm Sensing Hub), compute, NPU images		UFS	LPDDR5

3.2 External boot devices

The following table lists the GPIOs for external boot.

Table 3-2 External boot devices

FAST_BOOT GPIO [3:0]	GPIO no.				Boot device
	90	76	47	27	
0000	0	0	0	0	Default: UFS0 → SDC2 → USB0 → eDL (USB0)
0001	0	0	0	1	SDC2 → UFS0 → eDL (USB0)
0010	0	0	1	0	SDC2 → eDL (USB0)
0011	0	0	1	1	USB0 → eDL (SDC2 → USB0)
0100	0	1	0	0	QSPI → eDL (SDC2 → USB0)
0101	0	1	0	1	SPI → eDL (SDC2 → USB0)
0110	0	1	1	0	UFS0 → USB0 → eDL (USB0)
0111	0	1	1	1	Same as Fastboot[3:0] = 0000 UFS0 → SDC2 → USB0 → eDL (USB0)
Others					Reserved

NOTE

- eDL – Emergency Download. Boot from SD is not supported if PM8250 boot option pin GPIO_05 is grounded. For details, see the *PM8150/PM8250 Device Specification* (80-PD995-1).
- The FASTBOOT[4] fuse in OEM_CONFIG region of QFPROM does not have a corresponding GPIO. See the *QRB5165 QFPROM Programming Reference Guide*

(80-PV086-97) for fuses related to FASTBOOT settings and their impact on the boot device selection and boot flow.

3.3 Boot configuration GPIO settings

The following table lists the boot configuration and GPIO settings.

Table 3-3 Boot configuration and GPIO settings

Boot configuration no.	GPIO no.	Function	Bit values
BOOT_CONFIG_0	GPIO_128	WDOG disable	1 – Disables WDOG 0 – Enables WDOG
BOOT_CONFIG_1	GPIO_27	Fastboot bit 0	Selects external boot device
BOOT_CONFIG_2	GPIO_47	Fastboot bit 1	
BOOT_CONFIG_3	GPIO_76	Fastboot bit 2	
BOOT_CONFIG_4	GPIO_90	Fastboot bit 3	
FORCE_USB_BOOT	GPIO_132	Force USB download	0 – Releases USB download mode 1 – Enables USB download mode

NOTE External pulls or logic high on BOOT_CONFIG [4:1] before RESOUT_N getting asserted high can force QRB5165 device to boot from an unintended boot device.

3.4 FORCE_USB_BOOT

During development or factory production, a boot from the primary USB 3.1 port can be forced by using GPIO_132. FORCE_USB_BOOT (GPIO_132) always takes precedence, regardless of the state of the BOOT_CONFIG GPIOs or FAST_BOOT_SEL fuses.

The FORCE_USB_BOOT is checked first during the boot device detection, prior to BOOT_CONFIG GPIO states. GPIO_132 = 1 forces the QRB5165 device to boot from the primary USB 3.1 port.

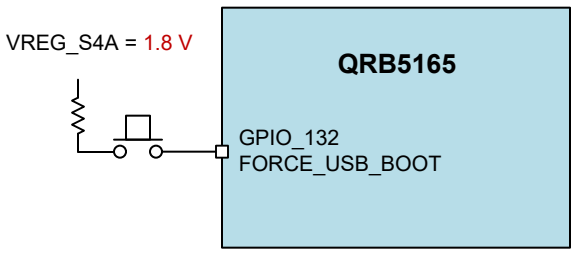


Figure 3-2 FORCE USB BOOT

3.5 Fastboot configuration

The boot configuration control is implemented within the security controller based on both the boot-related GPIOs and the boot-related QFPROM settings. The BOOT_CONFIG [4:1] settings determine the external boot devices, as shown in the following figure. The FAST_BOOT fuses (OEM_CONFIG_ROW0_LSB) can also select the same fastboot options.

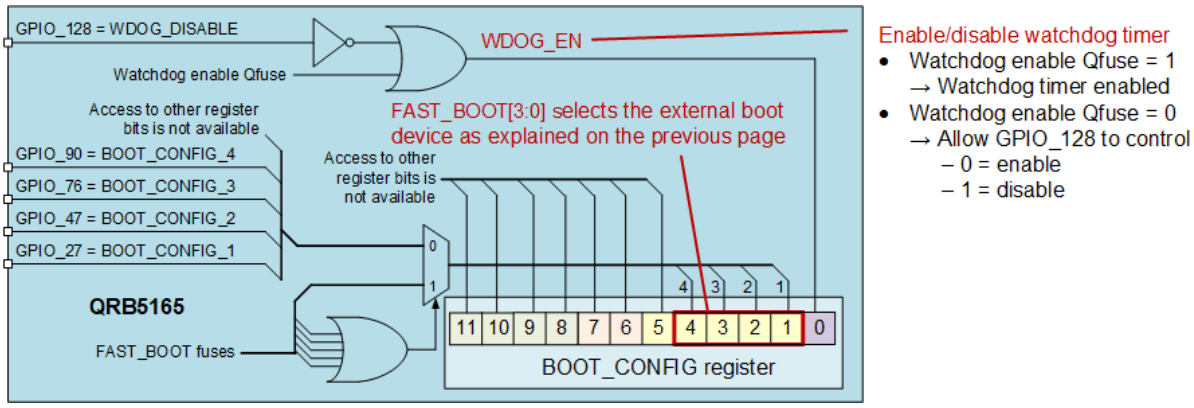


Figure 3-3 Fastboot configuratoin

3.6 Secure boot GPIO mapping

The secure boot ensures that the QTI code and the OEM code cannot be modified by another entity. The secure boot configuration is application-based. The applications code creates a root of trust (ROT):

- It is responsible for validating the code image and for validating the boot code stored in external memory.
- It confirms the code originated from a trusted authority (authenticity) and verifies that the code is in its original form (integrity).
- The digital signatures validate the external code image and establish the system security level.
- The ROT verifies the code image version (compares the signed version label to the value in Qfuse).
- Version control ensures that an old revoked code image is unusable.

See the *QRB5165 QFPROM Programming Reference Guide* (80-PV086-97) for details.

Table 3-4 Supported external memory configurations

Fuse name	Bit no. in row	Blow value	Comments
OEM_SECURE_BOOT1_ROM_PK_HASH_IDX0	0	0	If PK_HASH_IN_FUSE = 0, this value selects which root certificate hash to use from the ROM table.
OEM_SECURE_BOOT1_ROM_PK_HASH_IDX1	1	0	
OEM_SECURE_BOOT1_ROM_PK_HASH_IDX2	2	0	

Table 3-4 Supported external memory configurations (cont.)

Fuse name	Bit no. in row	Blow value	Comments
OEM_SECURE_BOOT1_ROM_PK_HASH_IDX3	3	0	
OEM_SECURE_BOOT1_PK_HASH_IN_FUSE	4	0	For boot configuration 1: ■ If this bit = 0, use the internal ROM hash index and OEM_SECURE_BOOT1_ROM_PK_HASH_IDX[3:0] for the root certificate hash. ■ If this bit = 1, use the value stored in OEM_PK_HASH for the root certificate hash.
OEM_SECURE_BOOT1_AUTH_EN	5	1	Blow this bit to enable secure boot for applications and other peripheral images. When this bit = 1, it enables authentication for any code that references secure boot configuration 1.
OEM_SECURE_BOOT1_USE_SERIAL_NUM	6	0	If this bit = 1, then the unique device serial number is required in the code authentication for boot configuration 1.

4 SoC subsystems

4.1 Processor

4.1.1 Kryo 585 application processor

The features of Kryo 585 application processor are listed below:

- Customized 64-bit Arm® v8.2-compliant octa-core Kryo applications processor
- Eight Kryo cores organized into two clusters:
  - Four high-performance Kryo cores – Gold cluster
    - Three Gold cores targeting 2.5 GHz
    - One Gold prime core targeting 2.8 GHz
  - Four low-power Kryo cores targeting 1.7 GHz for quad core – Silver cluster
- Dedicated L2 cache per Kryo Gold core and per Kryo Silver core
- L3 shared cache at 4 MB

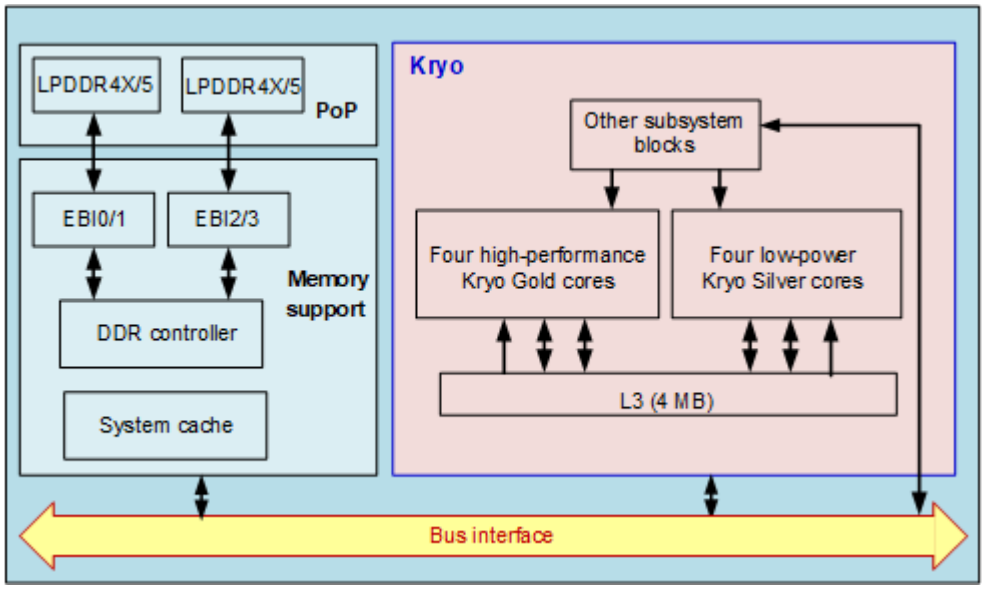


Figure 4-1 Kryo 585 application processor

4.1.2 Kryo 585 bus supply

The SMPS outputs S7A, S8A, and S9A are tied together at the PMIC and routed as a single trace to the Kryo Gold cluster. The SMPS output S10A is routed as a separate trace to Kryo Silver cluster.

Each Kryo core inside the QRB5165 device has a global distributed head switch (GDHS) to power the core.

Place the 3-terminal (3T) output bulk capacitors as close as possible to the QRB5165 device to reduce series resistance and ensure transient performance.

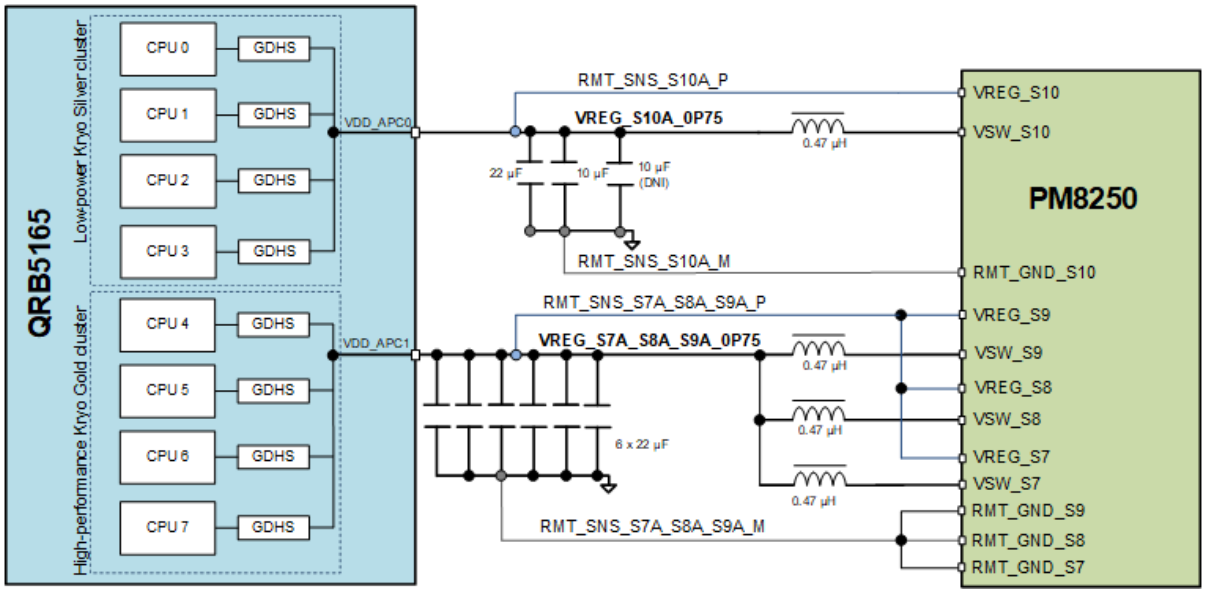


Figure 4-2 Kryo 585 power supply

4.1.3 Kryo clock and voltage controller

The high-performance (Gold) cluster and the low-power (Silver) cluster, along with the respective L2 cache, can operate on different clock sources at different frequencies.

- Within silver cluster, all four CPUs and L2 are synchronous.
- Within gold cluster, CPU7 (prime core) with its L2 has its own clock source and CPU4-6 with L2 are synchronous.

Shared 4 MB L3 cache (power supply shared with APC0, but the L3 frequency can be different than APC0).

During QRB5165 device boot, for the applications primary boot loader (PBL) to execute, only CPU0 is turned on.

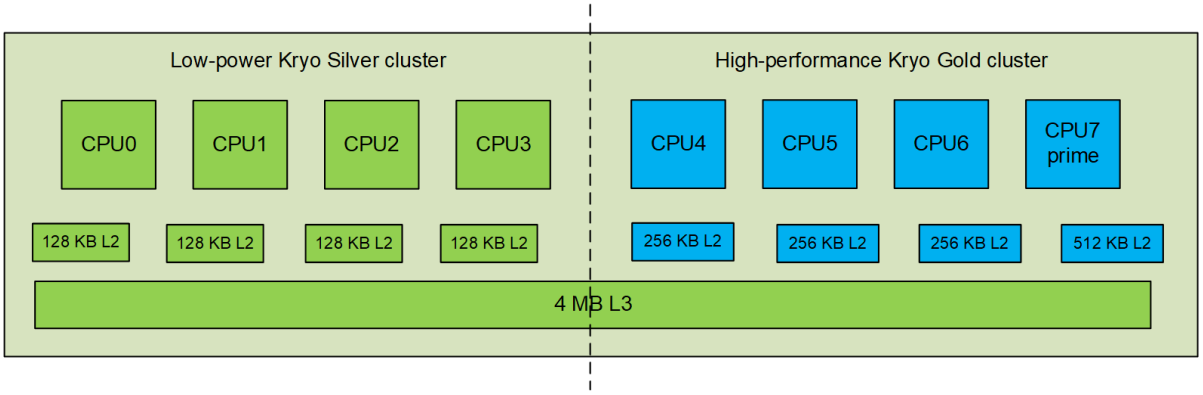


Figure 4-3 Kryo clock and voltage control

4.2 Memory

The QRB5165 supports internal, external, PoP, LPDDR memory as well as UFS storage.

4.2.1 Internal memory

The QRB5165 internal memory supports the following:

- A system cache (3 MB) is introduced to improve the performance of the memory system
  - Additive cache capacity
    - Provides 7 MB total cache (3 MB system cache + 4 MB Kryo L3 cache) for a single-thread workload
  - Allows better data sharing with other blocks
  - Improved QoS
- IMEM (256 KB) for system memory
  - This is the on-chip memory to reduce the dependency on LPDDR5
  - This is the inter-processor memory
  - This is also the secure memory for the TrustZone kernel
- GMEM (1.25 MB) dedicated for graphics use cases
- PIMEM (Pseudo-IMEM) an encrypted, integrity protected memory region
- Boot ROM
- Additionally, there are several local memories (SRAM) supporting each subsystem

4.2.2 External memory

The following table lists the QRB5165 supported external memory configurations and devices.

Table 4-1 QRB5165 supported external memory configurations

I/O port	Memory description
EBI	4 × 16 PoP LPDDR5 SDRAM with optional low power features up to 16 GB (see note)
UFS 3.1	Gear 4 two-lane

Following are few memories used on QTI QRB5165 + 4G/5G platforms:

- K3LK2K20BM-BGCN (6 GB LPDDR5)
- THGJFCT0T44BAIL (128 GB UFS3.0)
- KLUDG4UHDB-B2D1 (128 GB UFS3.0)

NOTE Maximum DRAM density of 16 GB is only from hardware capability and depends on memory supply/validation.

NOTE All validated DDR part numbers will be listed on Qualcomm website after the official announcement of the chipset passing CS (to view these, click the Hardware Components tab and select Memory).

4.2.3 PoP memory

The PoP technology uses an extra top substrate to accommodate area-array memory packages.

The QRB5165 device PoP supports 64-bit (4 × 16) 4-channel stacked memory LPDDR5, thereby increasing operational bandwidth and simplifying routing—no pre-stacking is needed. The technology includes a low-package warpage.

See the *PoP Memory for SM8250/SXR2130/QRB5165/QCS8250/QCS8245 Recommendations* (80-VP300-16) for more information.

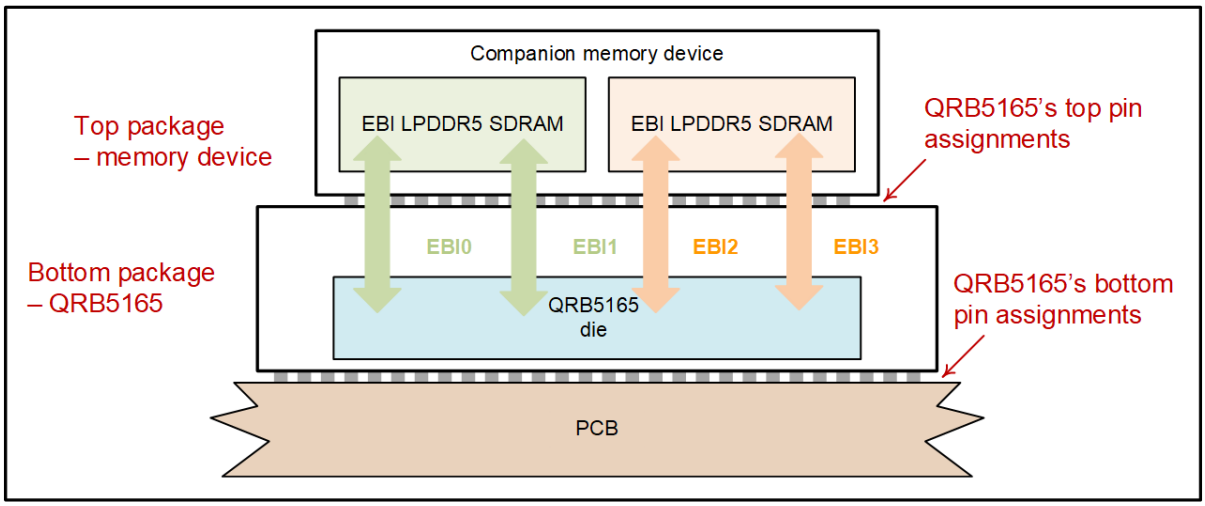


Figure 4-4 PoP memory

4.2.4 LPDDR memory

The following table shows the bottom BGA pin name for QRB5165-MPSP1099.

Table 4-2 QRB5165 pinmap – LPDDR5

QRB5165-LPDDR5 (MPSP1099)
NC
ZQ_A
NC
ZQ_D
VDD2H
VDD2L

A single PCB design supporting LPDDR5 is possible with proper BOM configuration as described below.

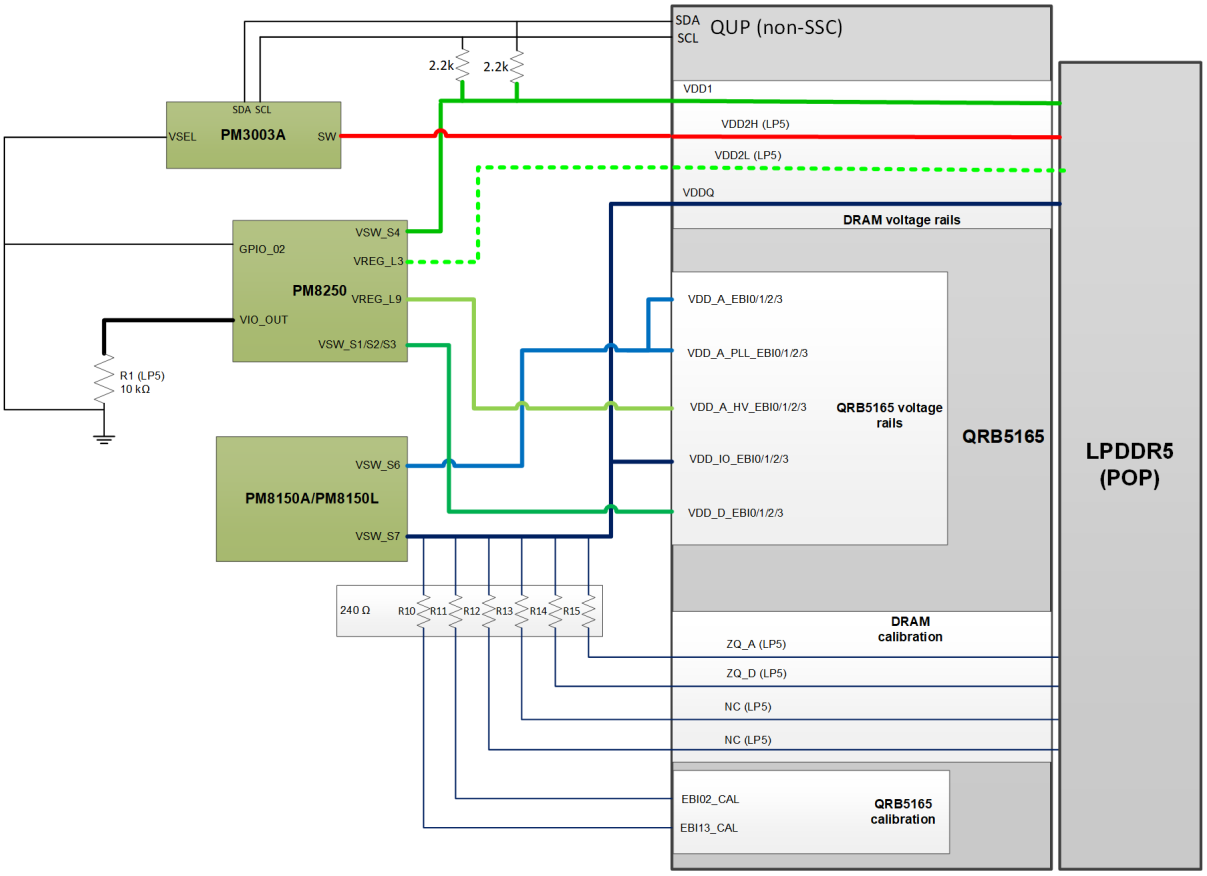


Figure 4-5 QRB5165 schematic – LPDDR5

Control of LPDDR5 – VDD2H

- Connect any unused QUP I^{2}C (non-SSC QUP) of QRB5165 dedicated to PM3003A. This is the recommended configuration.
- If all QUP I^{2}Cs (non-SSC QUP) are used, the I^{2}C of PM3003A can be shared on any QRB5165 QUP I^{2}C (whose master is TZ) with other peripheral devices.
- PM3003A I^{2}C (non-SSC QUP) can be shared on any QRB5165 QUP I^{2}C controlled by HLOS with other peripheral devices. This is the least recommended configuration.
- For the above three configurations, use 0 Ω series resistors, installed during development on PM3003A I^{2}C. These 0 Ω resistors need to be DNI'd in production to avoid malicious access.

4.2.5 Universal flash storage (UFS)

The UFS host controller is compliant with the following specifications:

- JEDEC specification for Universal Flash Storage (UFS) (JESD220C) – v3.0
- JEDEC specification for UFS Host Controller Interface (JESD223C) – v3.0
- MIPI Alliance specification for Unified Protocol (UniPro) – v1.8 (2-lane UFS0)/v1.6 (1-lane UFS1)
- MIPI Alliance specification for M-PHY – v4.1

NOTE QRB5165 provides one 2-lane UFS 3.1 port.

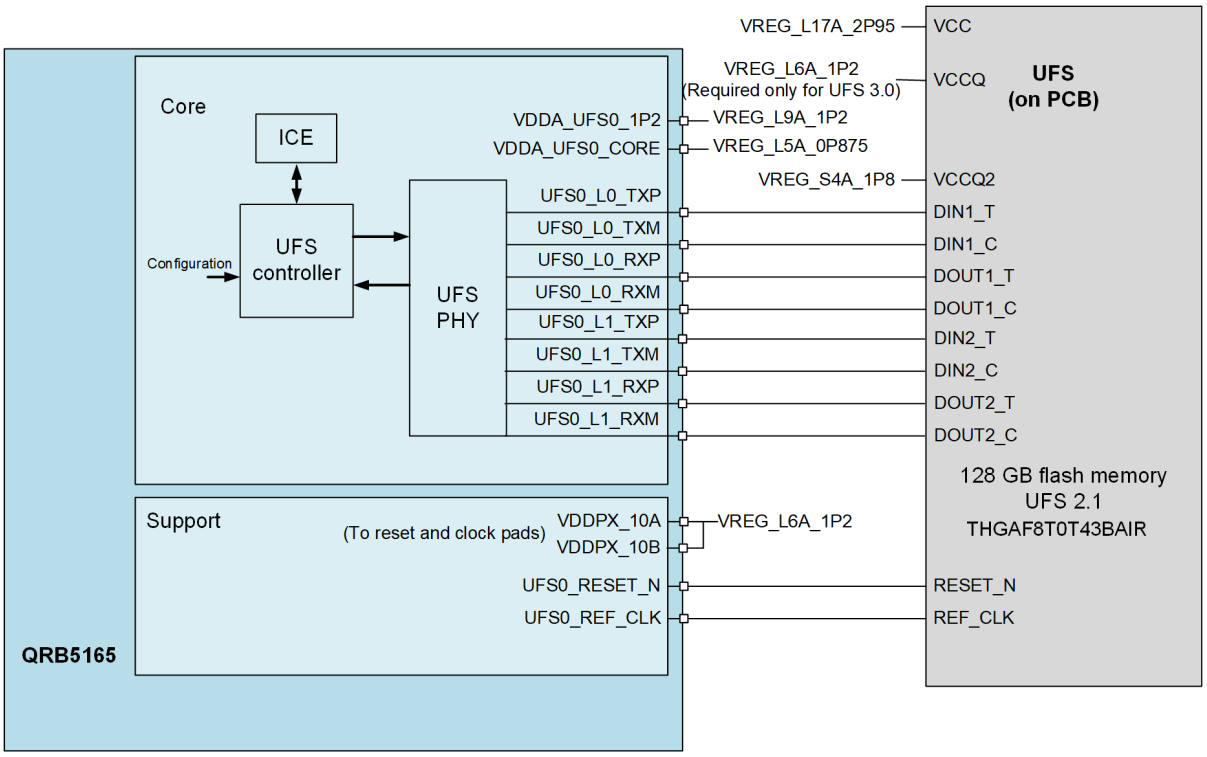


Figure 4-6 Universal flash storage

4.3 Multimedia

The following figure shows the QRB5165 multimedia subsystem.

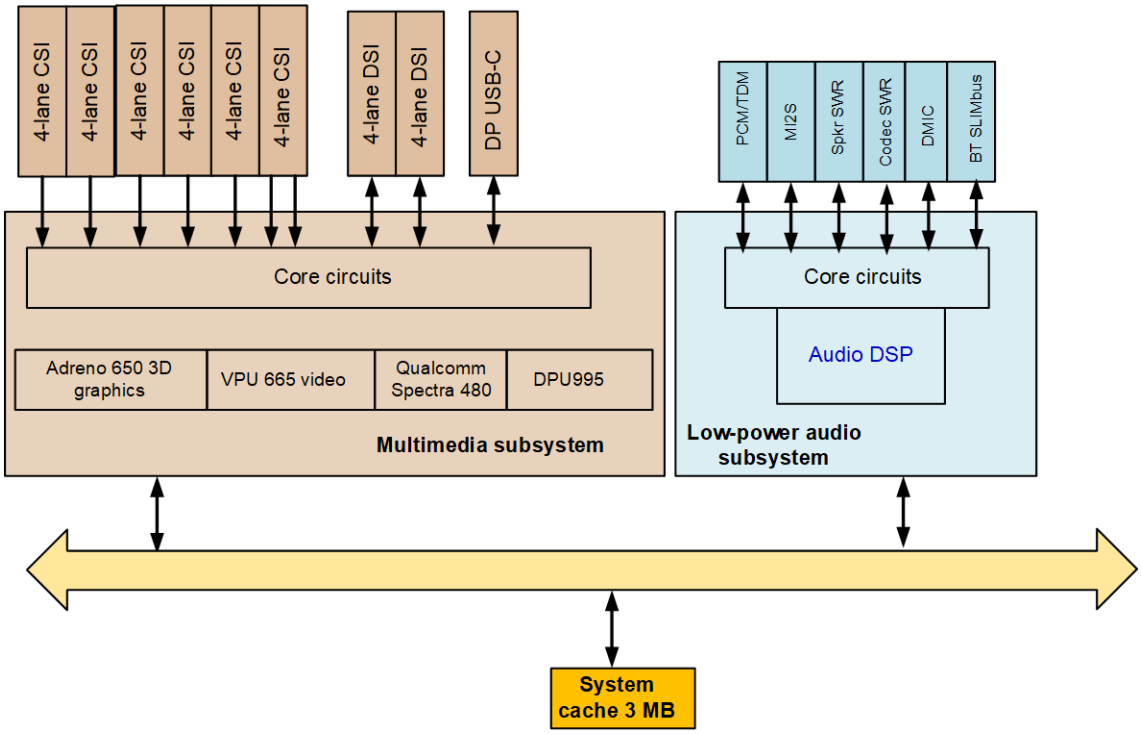


Figure 4-7 QRB5165 multimedia subsystem

4.3.1 Display

The display overview for QRB5165 is listed below.

- The QRB5165 chipset can support up to three 4K displays (one internal display through DSI and two external displays through DisplayPort).
- The device can also support 5040 × 2160 at 60 Hz, two 2560 × 2560 at 120 Hz for dual isplays, up to 30 bpp 4096 buffer width and 16 hardware-layer composition.
- The subsystem has an integrated clock controller.
- Two 4-lane DSI D-PHY 1.2 or two 3-trio C-PHY 1.1 with VESA DSC v1.1.
- Supports integrated DisplayPort v1.4 at 8.1 Gbps/lane.
  - Concurrent 4K60 DisplayPort and USB 3.0 operation over USB 3.1 Type-C.
- Multistream transport supports up to two 4K60, with two video streams, one audio on any one stream.
- Power improvements through inline rotation using system cache, Qualcomm Snapdragon Universal Bandwidth Compression 4.0, and RPMh.

4.3.2 Camera

Qualcomm Spectra 480 camera ISP

The QRB5165 device uses the Qualcomm Spectra 480 camera architecture, which has six interfaces for external cameras – six 4-lane interfaces with CSI D-PHY 1.2 or C-PHY 1.2.

Each of the D-PHY and C-PHY has a 1.2 interface has a data rate of 2.5 Gbps/lane and 1.2 interface has a data rate of 10.26 Gbps per T (4.5 Gsps/T) respectively.

The QRB5165 device can be connected to six cameras (4 + 4 + 4 + 4 + 4 + 4) or up to 12 (D-PHY)/18 (C-PHY) cameras. It can support seven concurrent camera usages. In case of dual camera, sync is done through the VSYNC pin.

The QRB5165 camera architecture supports HVX streaming.

The Qualcomm Spectra 480 camera architecture can support a 25 MP + 25 MP + 2 MP + 2 MP + 2 MP + 2 MP + 2 MP ISP configuration.

- IFE0 – 25 MP
- IFE1 – 25 MP
- IFE_Lite0 – 2 MP raw interface
- IFE_Lite1 – 2 MP raw interface
- IFE_Lite2 – 2 MP raw interface
- IFE_Lite3 – 2 MP raw interface
- IFE_Lite4 – 2 MP raw interface

Up to seven concurrent cameras

Dedicated computer vision processor (CVP) located in VPU

- Offload fundamental CV tasks from CPU, Compute DSP, GPU, and NPU
- Lower overall use-case power and increase system performance

CSI differential and three-phase

The CSI differential and three-phase configuration are listed below:

- Six 4-lane CSI (4 + 4 + 4 + 4 + 4 + 4) D-PHY 1.2 at 2.5 Gbps per lane or six 3-trio C-PHY 1.2 at 30.78 Gbps.
- C-PHY 1.2 is supported on the QRB5165 chipset, which uses single-ended signaling with an embedded clock (three wires per trio).

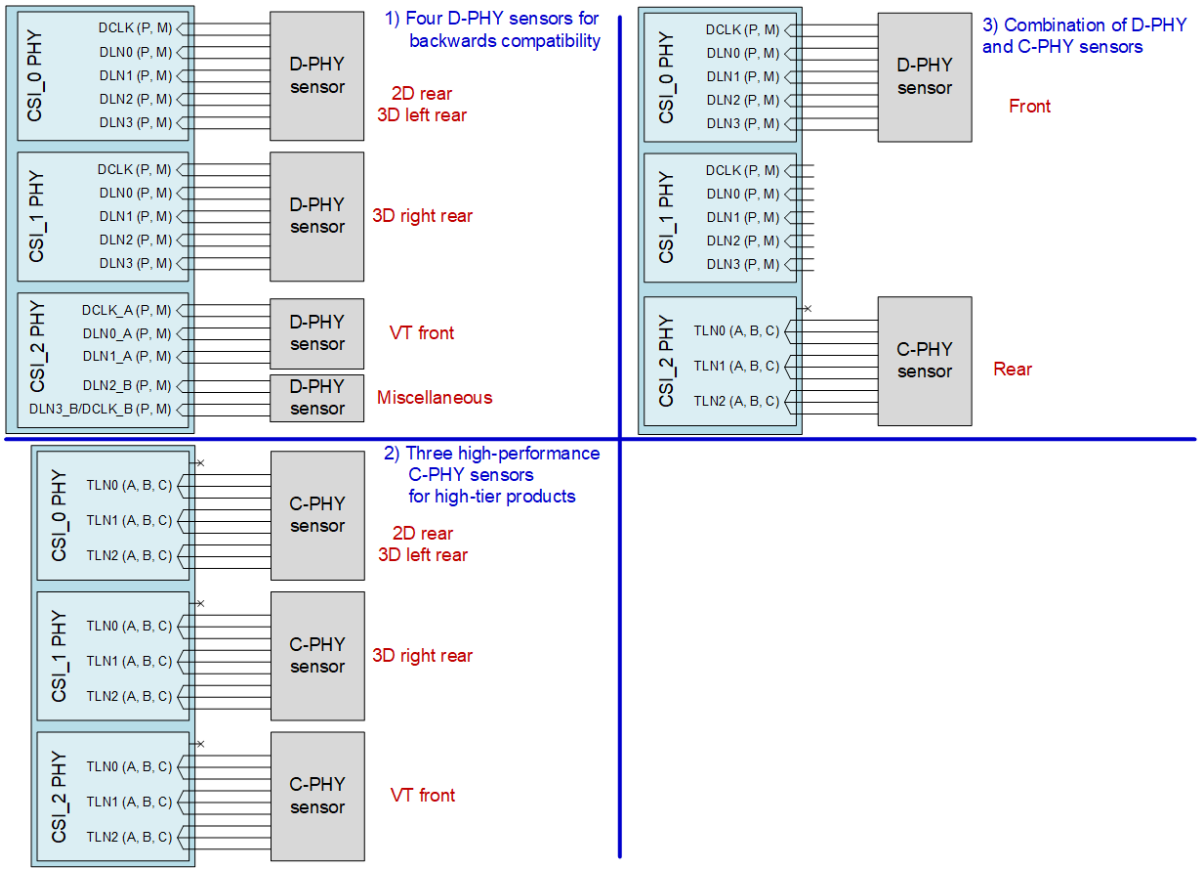


Figure 4-8 Camera support – example configurations for combination PHY

NOTE When a CSI PHY is used in a 2 + 1 dual-camera mode, only DLN3 can be used as CLK for the second camera.

NOTE These are only example configurations. Other configurations are possible.

Dedicated camera I²C ports for the camera control interface

In addition to the QUP I²C ports, there are four I²C ports that are dedicated to the camera control interface (CCI); do not use GPIO_101–GPIO_108 as general-purpose I²C ports.

Depending on the use case, the front and rear cameras can either share the CCI_I2C ports, or can have separate CCI_I2C ports.

It can support up to seven CAM_MCLK running at 19.2 MHz by default.

- CAM_MCLKx peak-to-peak jitter < 400 ps with default frequency.
- Contact QTI customer engineering before using any CAM_MCLK frequency other than 19.2 MHz, to investigate if jitter is acceptable for such cases.

Camera related GPIOs

The following table lists the camera related GPIOs.

Table 4-3 Camera related GPIOs

GPIO no.	Function	Comments
92	#CAM1_RST_N	Camera 1 reset
93	#CAM0_RST_N	Camera 0 reset
94	CAM_MCLK0	Sensor 0 master clock
95	CAM_MCLK1	Sensor 1 master clock
96	CAM_MCLK2	Sensor 2 master clock
97	CAM_MCLK3	Sensor 3 master clock
98	CAM_MCLK4	Sensor 4 master clock
99	CAM_MCLK5	Sensor 5 master clock
100	CAM_MCLK6	Sensor 6 master clock
101	CCI_I2C_SDA0	I^2C master 0 SDA
102	CCI_I2C_SCL0	I^2C master 0 SCL
103	CCI_I2C_SDA1	I^2C master 1 SDA
104	CCI_I2C_SCL1	I^2C master 1 SCL
105	CCI_I2C_SDA2	I^2C master 2 SDA
106	CCI_I2C_SCL2	I^2C master 2 SCL
107	CCI_I2C_SDA3	I^2C master 3 SDA
108	CCI_I2C_SCL3	I^2C master 3 SCL
109	CCI_TIMER0/#CAM3_RST_N	CCI timer/Cam 3 reset
110	CCI_TIMER1	CCI timer
111	CCI_TIMER2	CCI timer
112	CCI_TIMER3/CCI_ASYNC_IN1	CCI timer/camera interrupt
113	CCI_TIMER4/CCI_ASYNC_IN2	CCI timer/camera interrupt
114	CCI_ASYNC_IN0/#CAM6_RST_N	Camera interrupt/cam 6 reset

4.3.3 Audio

For details on QRB5165 LPASS, see the *SM8250/SXR2130P/QRB5165/QRB5165N/QCS8250/QCS8245 Audio Codec Hardware Overview Design Guidelines/Training Slides* (80-PL546-5C).

For design details, see the *Qualcomm Aqstic WCD9380/WCD9385 Audio Codec Hardware Design Guidelines/Training Slides* (80-PL335-5).

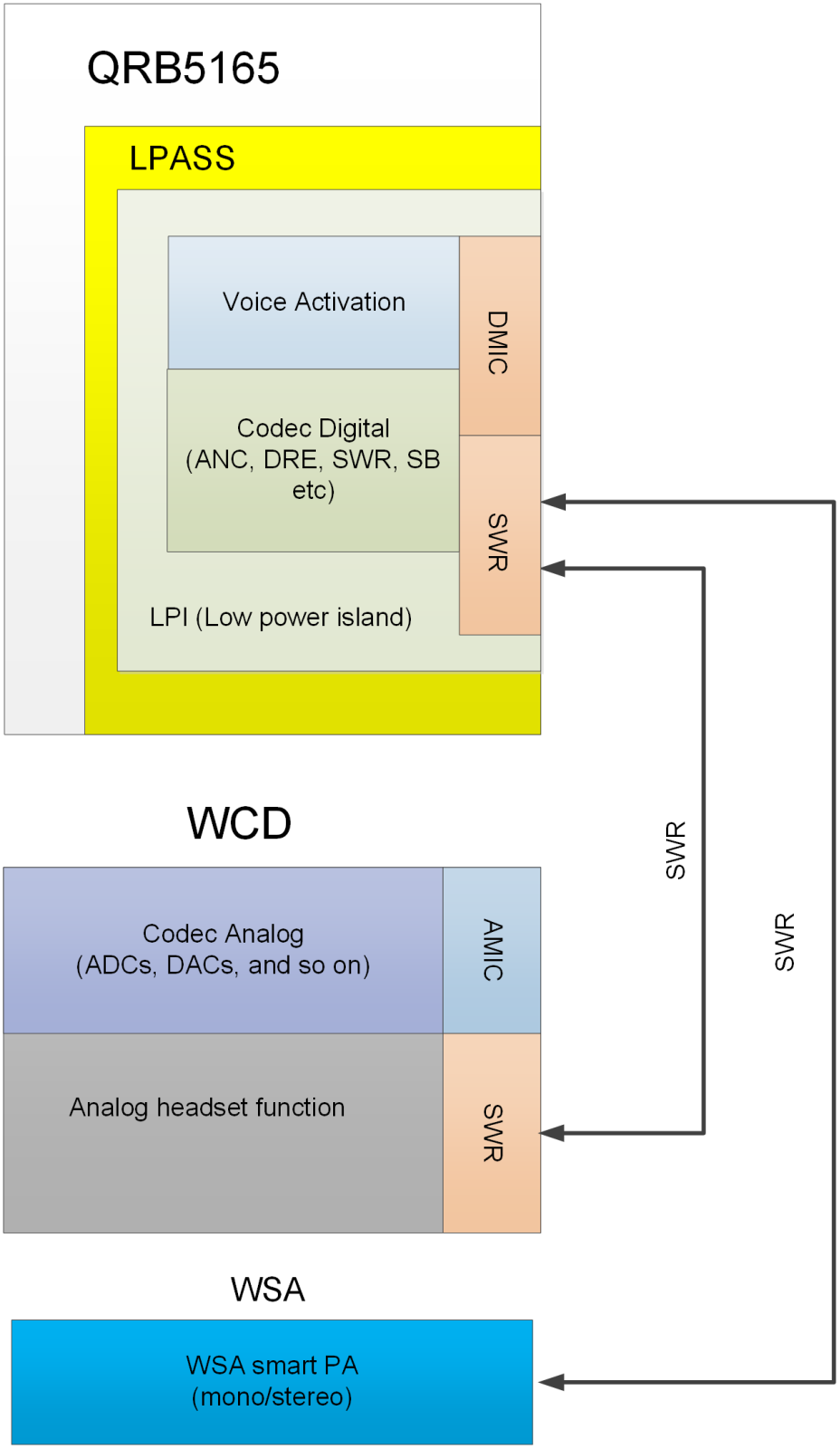


Figure 4-9 Audio interfaces

4.4 Qualcomm sensing hub

The QRB5165 device has an integrated sensor subsystem called the Qualcomm Sensing Hub, which is dedicated to support low-power, always-on use cases. The Qualcomm Sensing Hub has its own dedicated power rails – VDD_LPI_CX and VDD_LPI_MX. The sensor subsystem can be left powered on, even when the rest of the QRB5165 device is in sleep mode.

A low-power Hexagon DSP is used for processing in the Qualcomm Sensing Hub. The Qualcomm Hexagon DSP has a dedicated 1.5 MB L2/TCM cache.

The Qualcomm Sensing Hub has dedicated I/Os (SSC_0–SSC_19 muxed behind GPIO_160 to GPIO_179) to communicate with the sensors. There are eight QUP cores in the Qualcomm Sensing Hub. The I/Os can support UART, I^2C, I3C, and SPI interfaces.

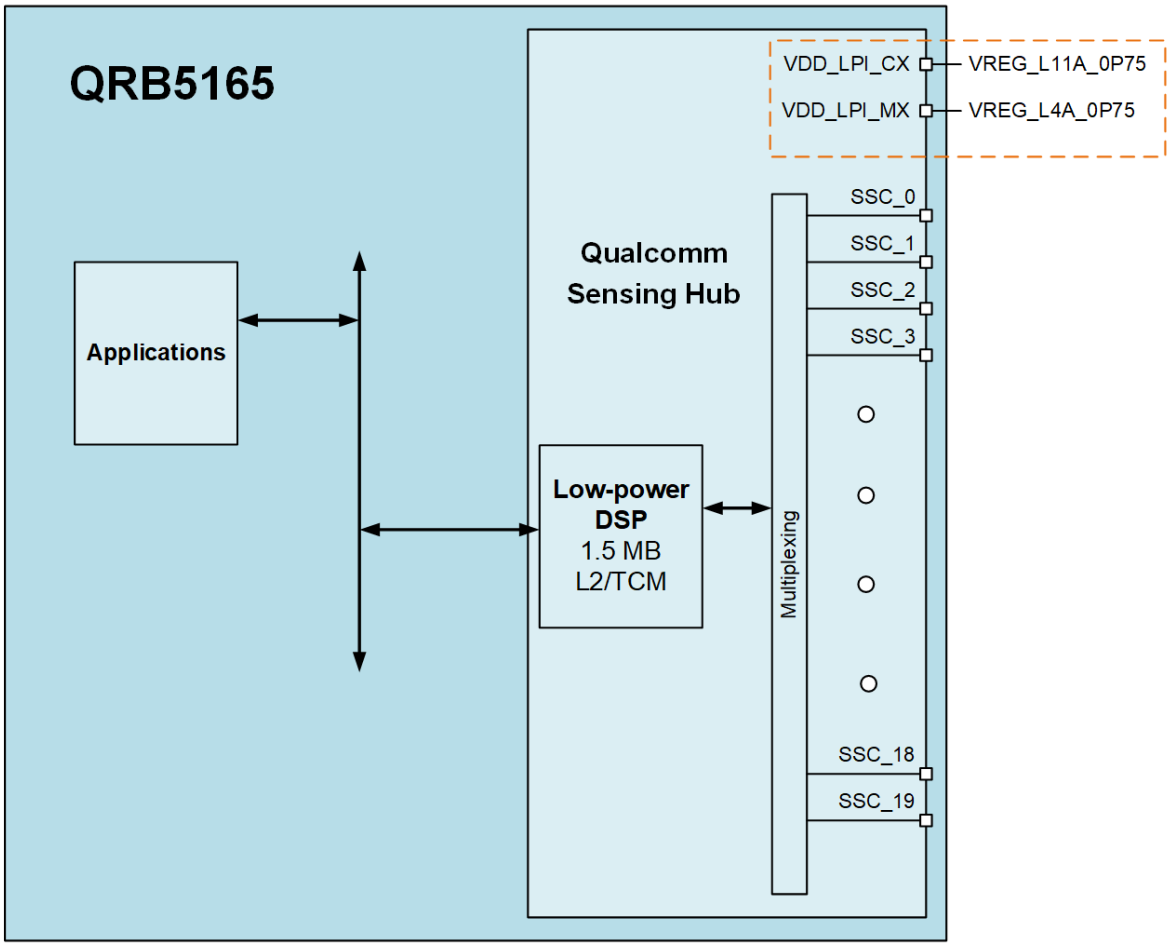
The Qualcomm Sensing Hub has direct access to the internal cores to support various sensor use cases. When GPIO_160 – GPIO_179 are configured as SSC_0–SSC_19, they cannot be used for non-Qualcomm Sensing Hub use cases. They are dedicated for use by the Qualcomm Sensing Hub Hexagon DSP.

For sensor interrupts, any available MPM-capable GPIO can be routed as an interrupt to the Qualcomm Sensing Hub subsystem using the TLMM registers:

- TLMM_WEST_MPM_WAKEUP_INT_EN_0
- TLMM_NORTH_MPM_WAKEUP_INT_EN_0
- TLMM_NORTH_MPM_WAKEUP_INT_EN_1
- TLMM_SOUTH_MPM_WAKEUP_INT_EN_0

NOTE Any available MPM GPIOs can be routed to the Qualcomm Sensing Hub. See the related software documents or file a case in Salesforce to the software team for detailed code changes.

VDD_LPI_CX and VDD_LPI_MX pins MUST be connected to the respective LDOs, even if the Qualcomm Sensing Hub is not used.



* The sensor core I/Os are now powered by VDD_PX3.

Figure 4-10 Qualcomm Sensing Hub

4.4.1 SSC_QUP MTP configuration

This is one of the sensor configurations that will be supported by QTI. The sensors connected to SSC0_1 (I3C), accel/gyro and magnetometer use DAE (data acquisition engine). The interrupt pins should be reserved for accel and gyroscope sensors even for I3C. The I3C IBI support on QRB5165 will not be commercialized. See the *QRB5165 Linux Ubuntu Software Reference Manual* (80-PV086-100) for the latest information.The MTP has dual accel/gyro connected to the same I3C bus.

Table 4-4 SSC_QUP MTP configuration

GPIO number	SSC port	SSC QUP assignment			MTP config (with I3C)	
		1st	2nd	3rd	Interface	Sensors connected
GPIO_160	SSC0	QUP0_0			I3C	Accel/Gyro (DAE) Mag (DAE) Dual Accel/gyro on MTP
GPIO_161	SSC1	QUP0_1				
GPIO_162	SSC2	QUP1_0	QUP5_4		I3C	Reserved for OIS
GPIO_163	SSC3	QUP1_1	QUP5_5			
GPIO_164	SSC4	QUP2_0			SPI	Not used (reserved for non-I3C config for SPI)
GPIO_165	SSC5	QUP2_1				
GPIO_166	SSC6	QUP2_2	QUP4_2			
GPIO_167	SSC7	QUP2_3	QUP4_3			
GPIO_168	SSC8	QUP3_0	QUP7_0	QUP2_4	I^2C	Reserved for OIS
GPIO_169	SSC9	QUP3_1	QUP7_1			
GPIO_170	SSC10	QUP4_0			I^2C	Pressure ALS/ Proximity SAR (capacitive) Humidity
GPIO_171	SSC11	QUP4_1				
GPIO_172	SSC12	QUP5_0			SPI	Reserved for WiGig
GPIO_173	SSC13	QUP5_1				
GPIO_174	SSC14	QUP5_2				
GPIO_175	SSC15	QUP5_3				
GPIO_176	SSC16	QUP6_2	QUP6_0		UART	BLE
GPIO_177	SSC17	QUP6_3	QUP6_1			
GPIO_178	SSC18	QUP7_2	QUP1_2		UART	WLAN
GPIO_179	SSC19	QUP7_3	QUP1_3			

4.4.2 SSC_QUP QRD configuration

The SCC_QUP QRD is the second configuration that will be supported by QTI. For DAE (data acquisition engine) support in this configuration, accel/gyro should be connected to SSC_4_7 (SPI) and magnetometer on a dedicated I^2C bus SSC_0_1. Other sensors should not share SSC_0_1. See the *QRB5165 Linux Ubuntu Software Reference Manual* (80-PV086-100) for more information.

Table 4-5 SSC_QUP QRD configuration

GPIO number	SSC port	SSC QUP assignment			QRD config (with I3C)	
		1st	2nd	3rd	Interface	Sensors connected
GPIO_160	SSC0	QUP0_0			I3C	Mag (DAE)
GPIO_161	SSC1	QUP0_1				
GPIO_162	SSC2	QUP1_0	QUP5_4		I3C	Reserved for OIS
GPIO_163	SSC3	QUP1_1	QUP5_5			
GPIO_164	SSC4	QUP2_0			SPI	Accel/Gyro (DAE)
GPIO_165	SSC5	QUP2_1				
GPIO_166	SSC6	QUP2_2	QUP4_2			
GPIO_167	SSC7	QUP2_3	QUP4_3			
GPIO_168	SSC8	QUP3_0	QUP7_0	QUP2_4	I^2C	Reserved for OIS
GPIO_169	SSC9	QUP3_1	QUP7_1			
GPIO_170	SSC10	QUP4_0			I^2C	Pressure
GPIO_171	SSC11	QUP4_1				ALS/ Proximity SAR (capacitive)
GPIO_172	SSC12	QUP5_0			SPI	Reserved for WiGig
GPIO_173	SSC13	QUP5_1				
GPIO_174	SSC14	QUP5_2				
GPIO_175	SSC15	QUP5_3				
GPIO_176	SSC16	QUP6_2	QUP6_0		UART	BLE
GPIO_177	SSC17	QUP6_3	QUP6_1			
GPIO_178	SSC18	QUP7_2	QUP1_2		UART	WLAN
GPIO_179	SSC19	QUP7_3	QUP1_3			

4.5 Always-on subsystem

The Always-on subsystem is responsible for the shared clock, voltage, and DDR management. The hardware-based resource and power management (RPMh) for faster response times–faster operating point transitions and enables faster transition to low-power modes. The Always-on processor handles the complex situations and dependencies.

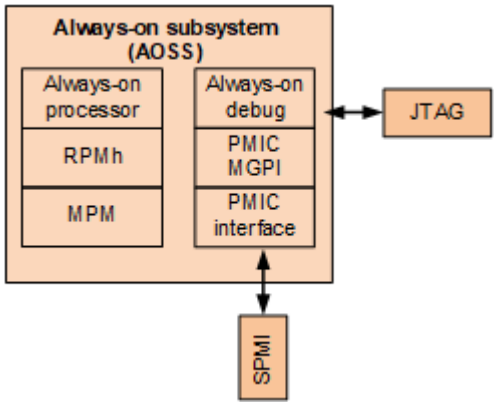


Figure 4-11 QRB5165 Always-on subsystem

Modem GRFC PMIC interface

The PMIC_MGPI (modem GRFC PMIC interface) reduces PMIC pins and routing congestion. The routing peripheral chips' clock enables signals to the QRB5165 MGPI_GPIOs. It also enables more system power control flexibility. The following steps explain the working of PMIC_MGPI:

1. A signal trigger from an internal QRB5165 signal or MGPI_GPIO is sent to QRB5165 PMIC_MGPI block.
2. The PMIC_MGPI trigger (rising or falling) corresponds to an 8-bit data command to a preconfigured SPMI SID/GID address.
3. The PMIC_MGPI block receiving the trigger sends an interrupt to the PMIC arbiter to enable the clock and send the SPMI command (data + address).

Table 4-6 PMIC_MGPI and GPIO

MGPI number	MGPI source at QRB5165 level	Default use case
MGPI0	GPIO_11	
	GPIO_15	
	GPIO_136	
	GPIO_3	
	GPIO_27	
MGPI1	GPIO_7	NFC
	GPIO_100	
	GPIO_137	
	GPIO_39	
	GPIO_47	

Table 4-6 PMIC_MGPI and GPIO (cont.)

MGPI number	MGPI source at QRB5165 level	Default use case
	GPIO_132	
MGPI2	GPIO_14	WiGig
	GPIO_68	
	GPIO_138	
	GPIO_51	
	GPIO_1	
	GPIO_109	
MGPI3	GPIO_124	WLAN
	GPIO_112	
	GPIO_133	
	GPIO_63	
	GPIO_70	
	GPIO_110	
MGPI4	GPIO_77	SD card detect
	GPIO_113	
	GPIO_134	
	GPIO_128	
	GPIO_129	
	GPIO_11	
MGPI5	NA	NA
MGPI6	Internal	Reserved
MGPI7	Internal	EUD

NOTE GPIO_7, GPIO_14, GPIO_124, and GPIO_77 are configured for NFC, WiGig, WLAN/ Bluetooth, and SD card detect interrupts by software default. They are configurable through software, if needed.

NOTE Within one MPGI, only one of the sources as mentioned in the table can be active at a time.

4.6 Qualcomm secure processing unit

The SPU240 is a standalone Qualcomm Secure Processing Unit with an embedded processor core, RAM, and integrated cryptography unit. It incorporates standalone boot capability and hardware-based anti-replay technology. It meets the Common Criteria's EAL4+ and FIPS 140-2 requirements.The SPU240 provides a secure, tamper-resistant execution environment. It is resistant to attacks such as:

- Side-channel analysis
- Fault attacks
- Invasive attacks

Following are some of the use cases that can benefit from the SPU:

Key management use cases: device unlocking, user data file encryption, and application usage of OS key management such as Keymaster

Biometrics use cases: ability to run core portions of biometric authenticators on the SPU while additional code runs on the TEE

Connected use cases: integration of use cases such as integrated UICC for eSIM

The specifications of SPU240 are:

- New crypto management unit (CMU)
- Independent boot chain (PBL)
- New key management unit
- CPU performance improvement
- Physical protection through multiple countermeasures
- RPM timers + watchdog
- Operating condition sensors

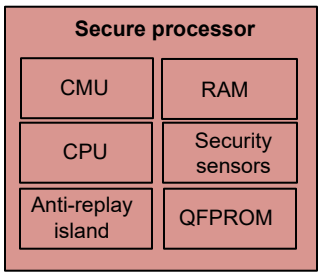


Figure 4-12 Secure processing unit

See the *Secure Processor Unit (SPU) – Anti-Replay Island (ARI) Overview for SM8150/SM8250/ SXR2130P/SA8155/QCS8250/QCS8245 Application Note* (80-PD867-16).

4.7 Compute DSP with Quad HVX and HCP 1.1

The features of Compute DSP are listed below.

- Hexagon DSP with Quad-HVX
  - Dual cluster base core for twice the scalar capability
  - Dual-HVX → Quad-HVX for twice the wide vector capability
    - HVX are new wide-vector instructions and registers to support high-performance vision processing and machine learning processing
- HCP1.1 (Hexagon co-processors)
  - Vision and imaging hardware accelerators
    - Fixed function configurable hardware to offload and accelerate software algorithmic functions

- Designed to be used with the fully programmable Hexagon DSP cores
  - Combination of Hexagon DSP plus HCP offers high performance and low power solutions while maintaining software flexibility
- Snapdragon UBWC 4.0
  - Memory compression to reduce bandwidth requirements for better performance and lower power than Snapdragon UBWC 3.0.

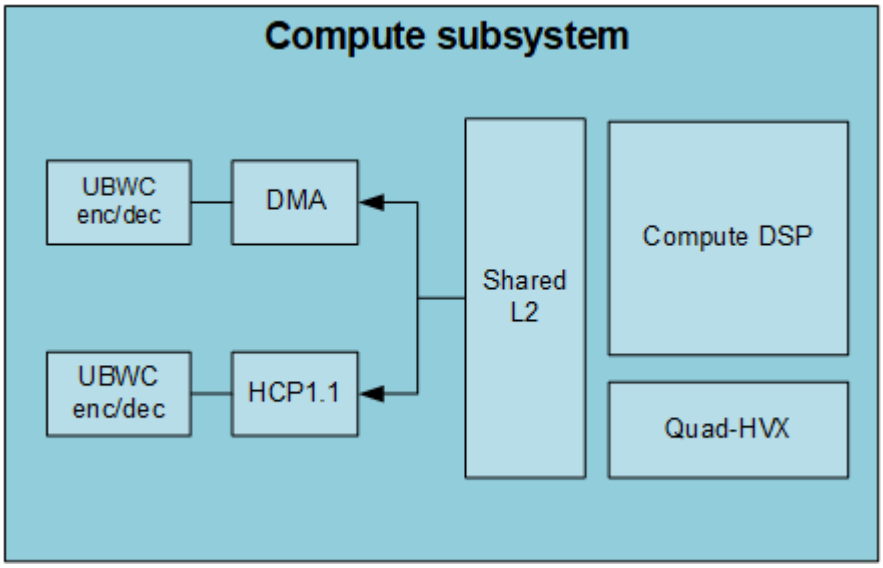


Figure 4-13 Compute DSP with Quad-HVX and HCP1.1

Use cases of HVX and HCP

The following table captures the use case benefits of HVX and HCP:

Table 4-7 Use cases of HVX and HCP

Use cases	Description
Video playback enhancement	■ Restoration of compressed video □ Codec noise reduction (CNR) – Edge artifact reduction (EAR) and so on □ Adaptive image enhancement – Content adaptive detail enhancement (CADE) – Local tone mapping (LTM) □ Frame rate conversion (FRC) – Slow motion playback, motion smoothing
Visual effects for social media	■ Augmented vision for immersive video
Computer vision (CV)	■ Qualcomm® Image Tracking ■ Scene classification

Table 4-7 Use cases of HVX and HCP (cont.)

Use cases	Description
Video capture enhancement	■ Low light video capture (such as Almalence SuperSensor or ArcSoft NightShot) ■ Image stabilization (such as ArcSoft Anti-Shaking stabilizer)
Machine learning	■ CNN (such as TensorFlow in Google Pixel) □ Object detection in images □ Facial recognition – Faces in general – Specific faces (such as girlfriend, boyfriend, spouse, children, and, so on) □ Scene classification – Image sorting/cataloging – Automatic camera setup □ LSTM (virtual assistants like Amazon Echo) – Command recognition – Voice recognition – Voice authentication □ Multiple other neural networks

4.8 Neural processing unit

The NPU230 is the dedicated neural processing unit for performance and always-on neural network (NN) use cases. The network execution engine consists of:

- 4096 8-bit and 1024 16-bit fixed point match – scalable for AI compute
- Generates complex address patterns to fetch data for MACs
- Native support for SNPE, Future Android NN, and Caffe 2 ONNX

It has a tightly coupled memory (256 KB) with built-in support for NPU-centric load/store operations and synchronization with the DMA. The NPU230 has a 3D DMA controller for DNN, which supports 3D data structures movement, streaming DMA, gather/scatter, padding/cropping in any dimension.

4.9 Connectivity

The following figure shows the connectivity of QRB5165.

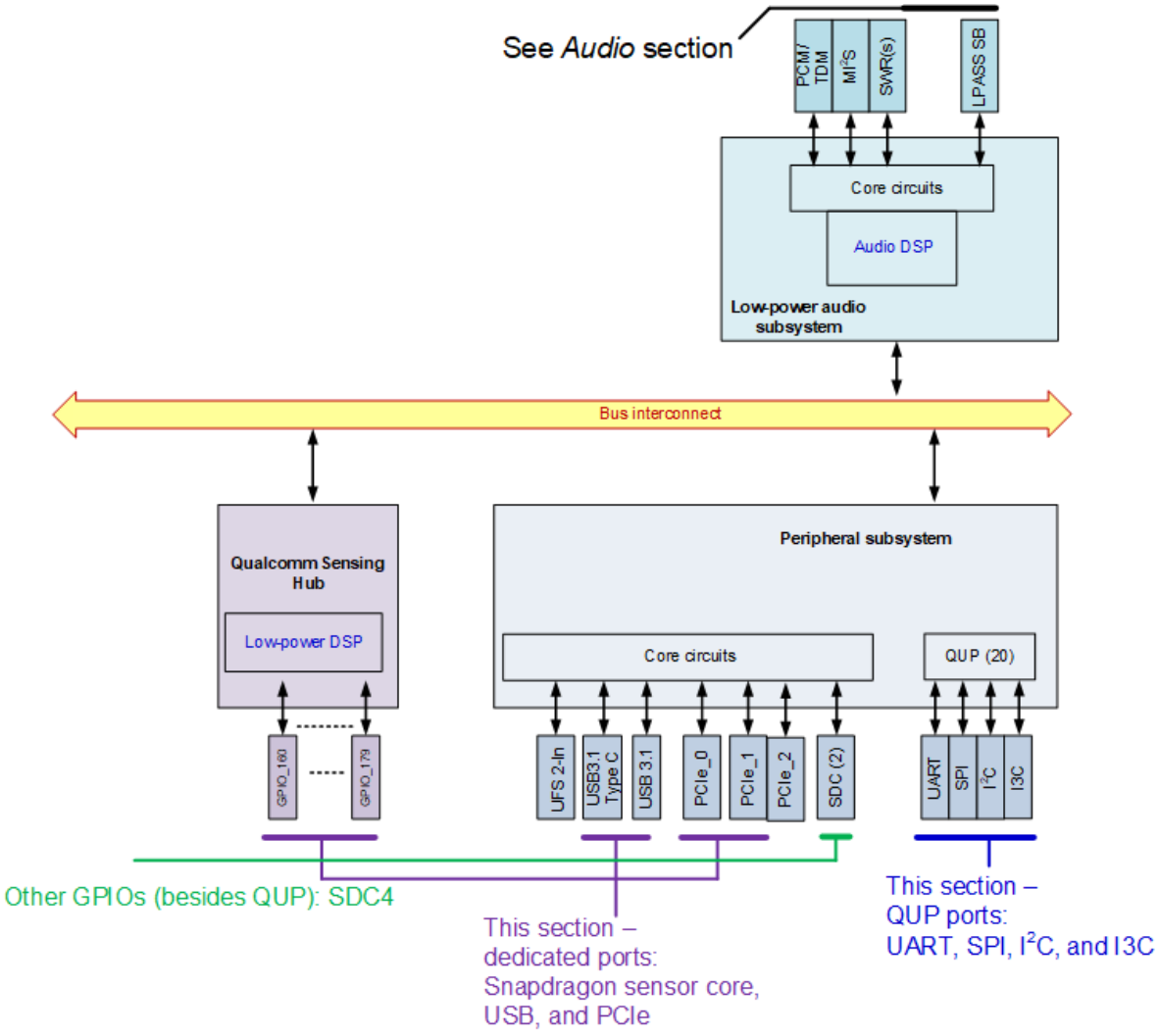


Figure 4-14 QRB5165 connectivity

4.9.1 Universal serial bus

The universal serial bus (USB) and the PCIe are the dedicated connectivity ports of QRB5165.

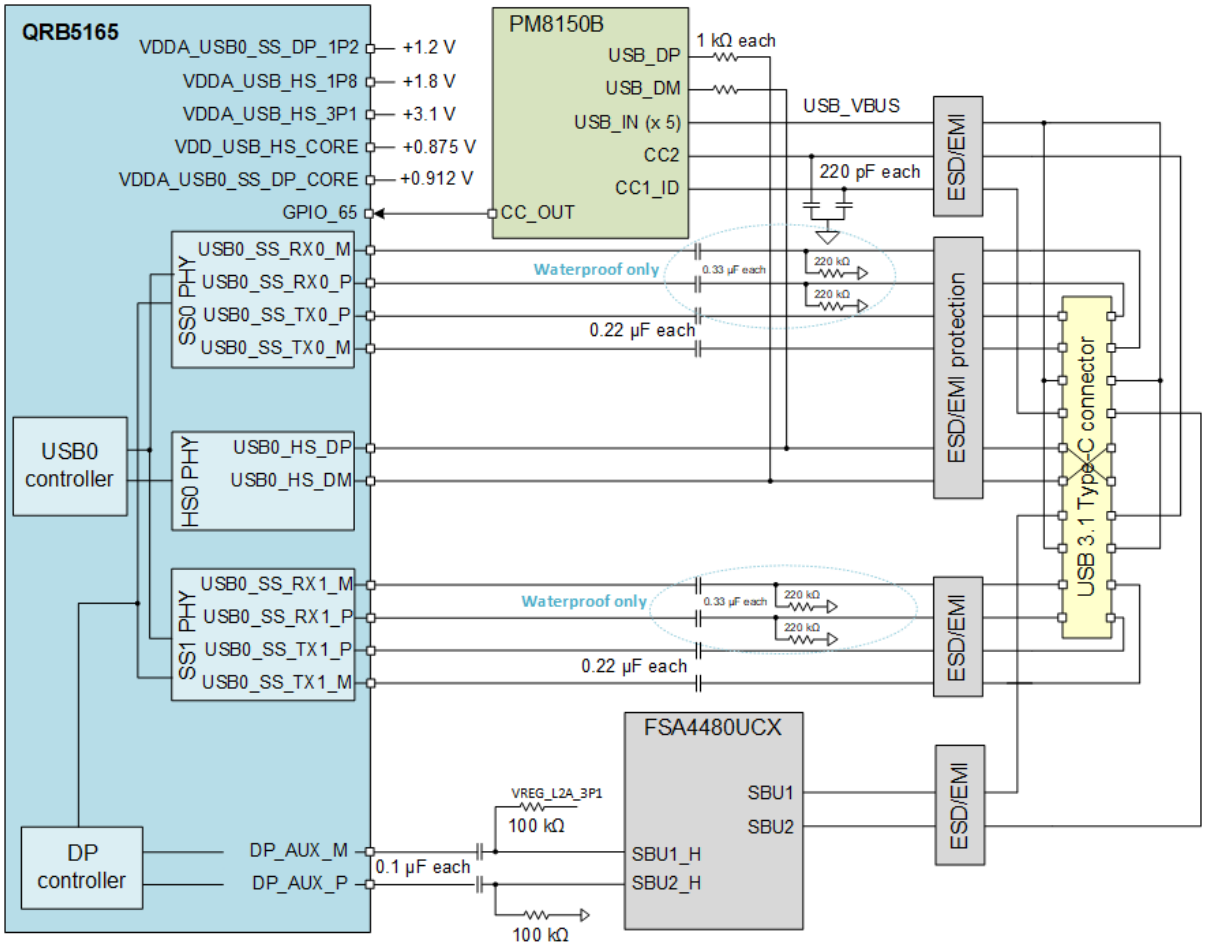


Figure 4-15 USB 3.1 Type-C

NOTE Due to the surge test (IEC 61000-4-5) carried out on USB 3.0 port, it is required that USB 3.0 PHY 0.9 V power supply (for example, VDDA_USB_SS_DP_CORE/ VDDA_USB_SS_CORE pins) is isolated or from a dedicated LDO.

The FSA4480UCX is an audio over USB switch which contains SBU switch. If audio over USB is not required, a different switch can be used to reduce BOM cost. For more detailed circuitry, see the *Design Package, QRD8250, Smart Phone, Electrical Hardware, PCB Schematic & Layout* (DP10-VL764-100A). The block diagram only shows the SBU switch for illustration purposes.

AUX_M/P must be disconnected in the system until the connection is switched into the DisplayPort mode, as per the USB Cable and Connector Specification. FSA4480UCX isolates the connection and switches AUX_M and AUX_P when needed.

100 kΩ pull-up and pull-down resistors on AUX_M and AUX_P are required, per the VESA DisplayPort Alt mode in the USB Type-C standard.

There is a secondary USB controller dedicated for USB 1.0 PHY which is not shown in the figure for simplification purposes.

Support USB 3.0 and DisplayPort concurrency mode

USB mode and DisplayPort mode can be simultaneously operating at USB 3.0 (10 Gbps) and DisplayPort (8.1 Gbps). In DisplayPort mode:

- QRB5165 can support up to 4-lane DisplayPort, 8.1 Gbps/lane.
- USB 2.0 can be supported simultaneously.
- USB 3.1 can be supported simultaneously.

	Port Select				
phy_mode	0	1	QMP lane	Type-c Receptacle pins	connector pin label
USB3 Only the 2 Tx/Rx pairs are assigned, but only one is in use	TX		lane_0_0	TX1+/-	A2, A3
	RX		lane_0_1	RX1+/-	B11, B10
		TX	lane_1_0	TX2+/-	B2, B3
		RX	lane_1_1	RX2+/-	A11, A10
DP Only All 4 lanes as Tx are assigned to DP	DP ML2	DP ML1	lane_0_0	TX1+/-	A2, A3
	DP ML3	DP ML0	lane_0_1	RX1+/-	B11, B10
	DP ML1	DP ML2	lane_1_0	TX2+/-	B2, B3
	DP ML0	DP ML3	lane_1_1	RX2+/-	A11, A10
Concurrent DP/USB 2 lanes assigned to USB and 2 to DP	TX	DP ML1	lane_0_0	TX1+/-	A2, A3
	RX	DP ML0	lane_0_1	RX1+/-	B11, B10
	DP ML1	TX	lane_1_0	TX2+/-	B2, B3
	DP ML0	RX	lane_1_1	RX2+/-	A11, A10

USB Type-C Receptacle Pins

A1	A2	A3	A9	A8	A7	A6	A5	A4	A10	A11	A12
GND	TX1+	TX1-	VBUS	CC1	D+	D-	SBU1	VBUS	RX2-	RX2+	GND
GND	RX1+	RX1-	VBUS	SBU2	D-	D+	CC2	VBUS	TX2-	TX2+	GND
B12	B11	B10	B9	B8	B7	B6	B5	B4	B3	B2	B1

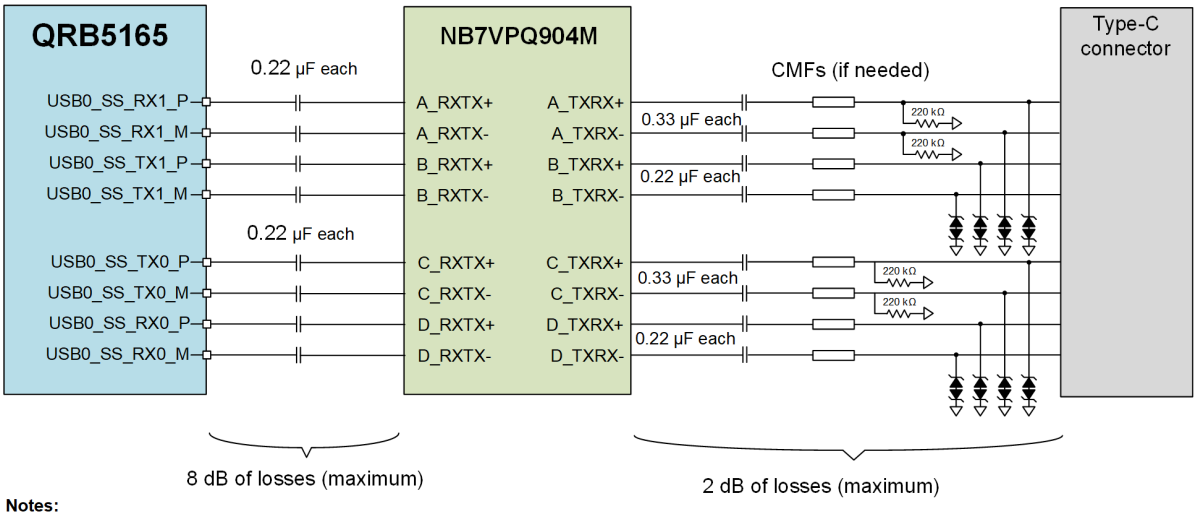
USB 3.1 and DispayPort re-driver recommendation

In order to support 10 Gbps USB 3.1 Gen 2, the insertion loss budget is very stringent due to various factors.

- PCB trace length
- Flex cable material
- ESD and common-mode choke filter

A re-driver is required if the following insertion loss budget cannot be met (for example, see the figure below)

- 10 Gbps USB 3.1 Gen 2 (+ 8.1 Gbps DisplayPort): -7 dB at 5 GHz
- 5 Gbps USB 3.1 Gen 1 (+ 8.1 Gbps DisplayPort): -5.5 dB at 2.5 GHz



Notes:
- Avoid using vias on the USB SS data lines.
- If using a CMF, it must be rated for USB SS Gen2 10 Gbps and be very low loss.
- If using a TVS diode, it must be extremely low capacitance, that is 0.25 pF or less.

Figure 4-16 Example of a re-driver circuit

4.9.2 PCIe

The QRB5165 chipset has three PCIe interfaces:

1. PCIe0 is a Gen 3 1-lane interface. It is used to connect to QCA639x (WLAN/Bluetooth IC) and the interface runs at Gen2 speed.
2. PCIe1 is a Gen 3 2-lane interface. It can be used for 8011ad (with QCA643x).
3. PCIe2 is a Gen 3 2-lane interface. PCIe2 is used to connect to 4G/5G modem IC.

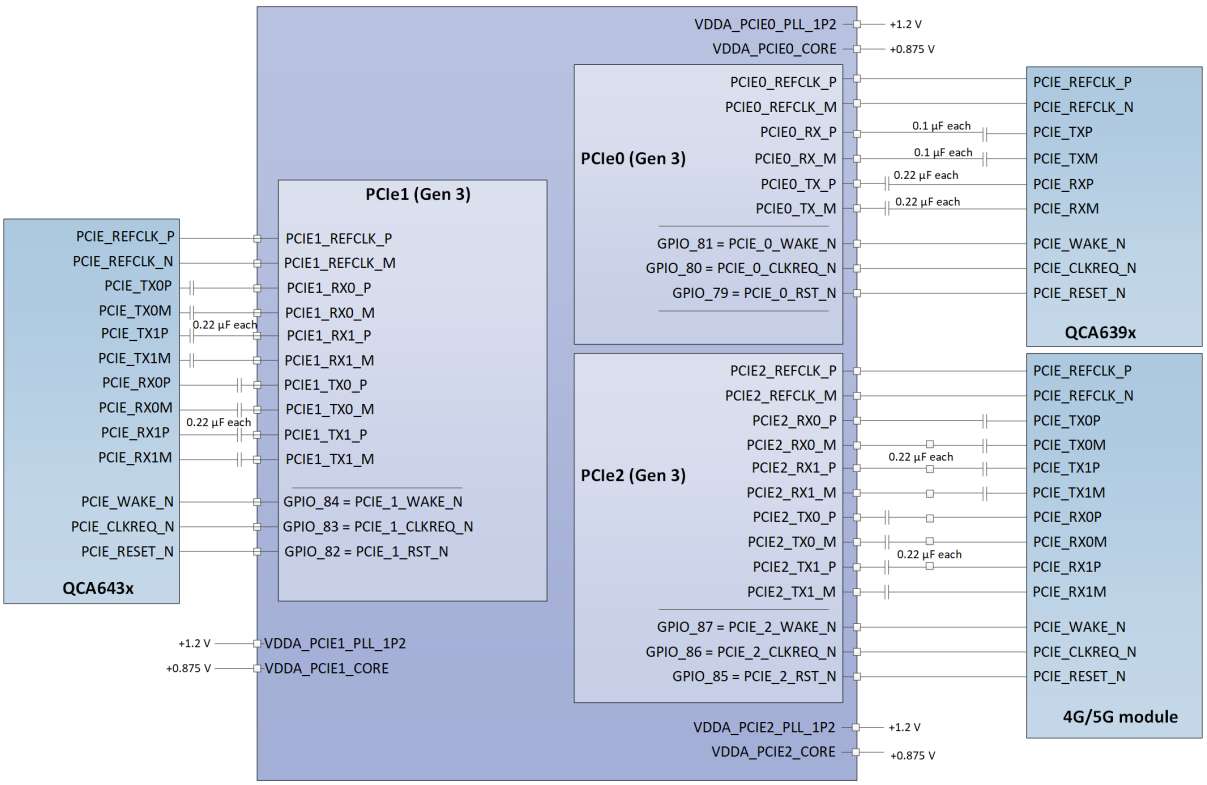


Figure 4-17 PCIe

NOTE External pull-ups are required for PCIE_WAKE_N and PCIE_CLKREQ_N signals. See the *Design Package, QRD8250, Smart Phone, Electrical Hardware, PCB Schematic & Layout* (DP10-VL764-100A) and *QCA6431 SIP for Qualcomm Snapdragon SM8150 Mobile Designs Reference Schematic* (80-YB966-47) for more information.

4.9.3 QUP ports (I^2C, I3C, SPI, and UART)

The QUP is highly flexible and programmable module for supporting a wide range of serial interfaces (I^2C, I3C, SPI, and UART). There are 20 QUP engines available in GPIOs and eight QUP engines available in Qualcomm Sensing Hub SSC_I/Os. Each QUP has up to seven lanes (I/O) numbered from 0 to 6. The supported assignments are listed in the table the for each QUP. Only one protocol can be selected in one QUP engine at a time. For example, simultaneous UART and I^2C functionality is no longer supported.

Table 4-8 QUP ports

Generic I/O	UART	I^2C	SPI	I3C
QUP_L0	CTS	SDA	MISO	SDA
QUP_L1	RFR	SCL	MOSI	SCL
QUP_L2	Tx	–	SCLK	–
QUP_L3	Rx	–	CS_0	–
QUP_L4	–	–	CS_1	–

Table 4-8 QUP ports (cont.)

Generic I/O	UART	I²C	SPI	I3C
QUP_L5	–	–	CS_2	–
QUP_L6	–	–	CS_3	–

NOTE See the *QRB5165 Data Sheet* (80-PV086-1) and the *QRB5165 Pin Assignment and the GPIO Configuration Spreadsheet* (80-PV086-1A) for detailed QUP assignments.

The serial interface can be configured using the following two steps:

1. **Pin configuration**: See the *QRB5165 Pin Assignment and GPIO Configuration Spreadsheet* (80-PV086-1A) and select the QUP I/O functions on GPIOs by writing to TLMM_GPIO_CFGn [FUNC_SEL] field. For example, if I²C is needed, only qup_l0 and qup_l1 should be configured on GPIOs. If a 4-wire SPI is needed, qup_l0 to qup_l3 should be configured on GPIOs.
2. **Protocol selection**: Select and configure the serial protocol engine by writing to the QUP registers.

I²C and I3C

I²C is a 2-wire bus that can be routed to multiple devices; each line of each bus is supplemented by a 2.2 kΩ pull-up resistor.

I3C is also a 2-wire bus interface with higher performance; each line of each bus does not require any pull-up resistor.

Each QUP core can be configured to support one I²C interface, always using lane 1 and lane 0. Only certain QUPs or SSC_QUPs support I3C interface.

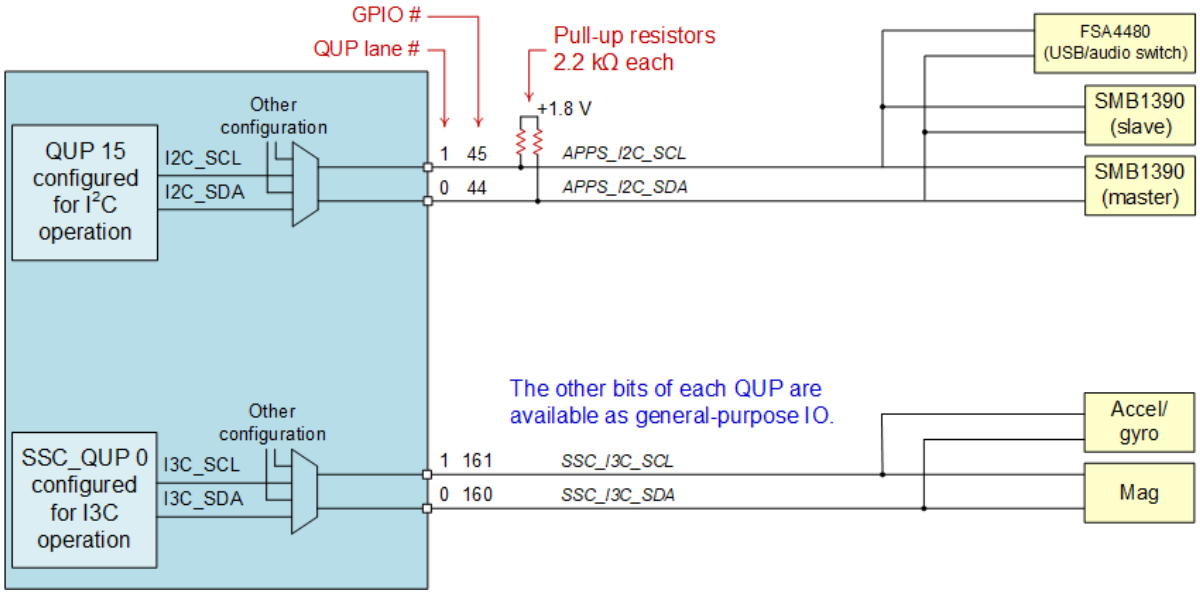


Figure 4-18 I²C and I3C configuration

SPI

The following figure shows the SPI configuration for QUP ports.

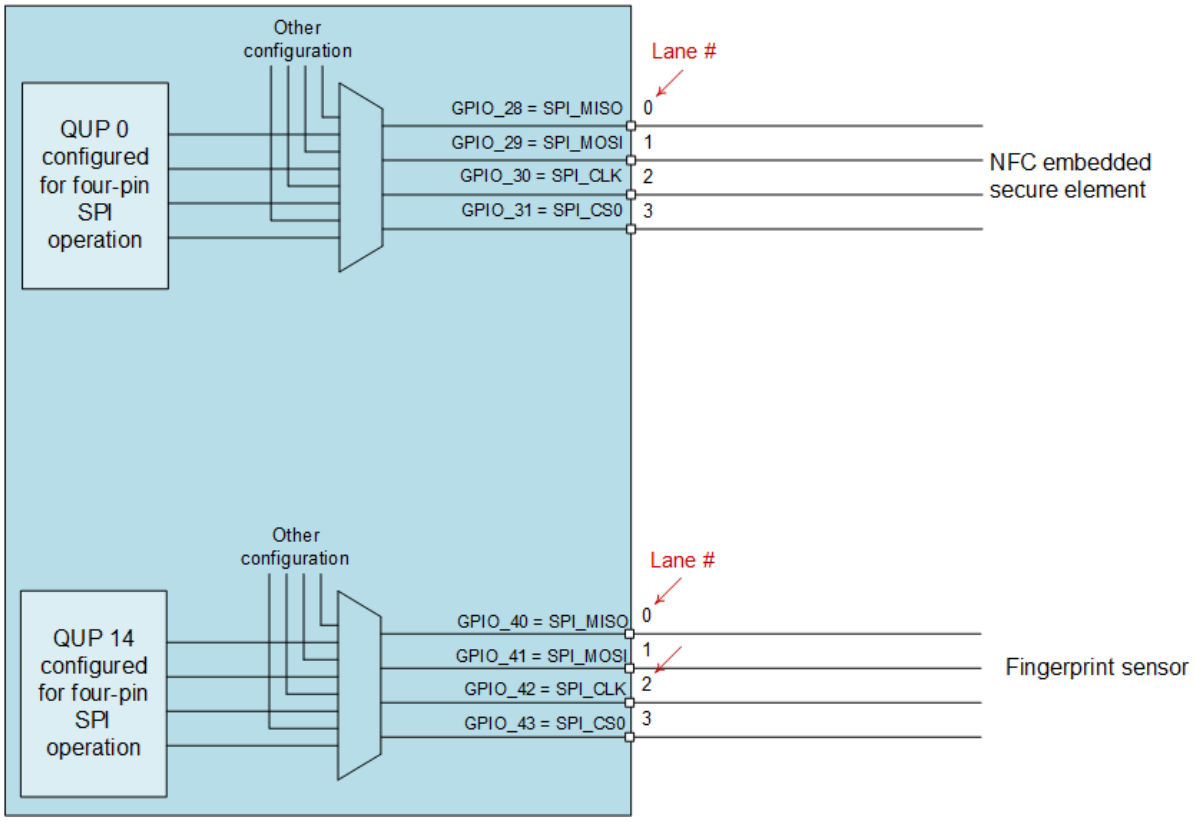


Figure 4-19 SPI configuration

4.9.4 Non-QUP GPIO ports – SDCs

There are two SDC interfaces:

- SDC2 has dedicated pins
- SDC4 is configurable via GPIOs

See the *MultiMediaCard/Secure Digital Card Application Note* (80-V7837-1) for more details.

Table 4-9 SDC specifications

SDC	Function	Width	Voltage	Maximum clock frequency	Supported modes
2	SD 3.0/MMC	4 bits	1.8 or 2.95 V	202 MHz	DS, HS, SDR12, SDR25, SDR50, SDR104, and DDR50
4	DMB	4 bits	1.8 V	100 MHz	SDR12, SDR25, SDR50, and DDR50

The QRB5165 device does not include SDC1 and SDC3 interfaces.

NOTE SDC4 has nonsequential GPIO assignments.

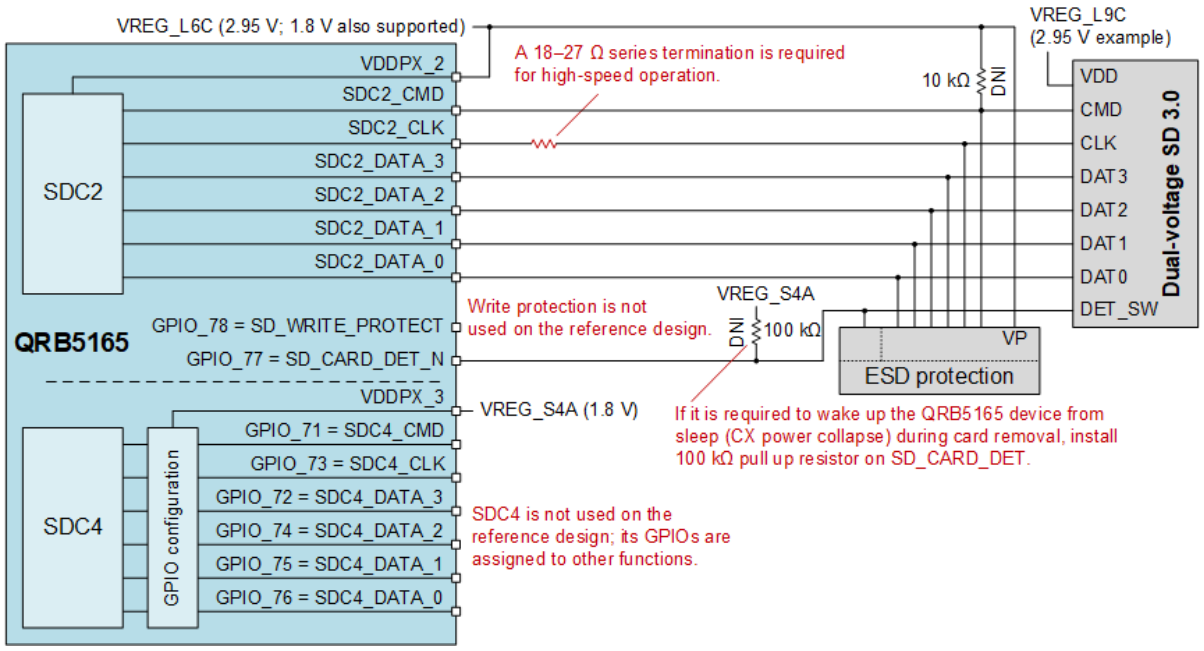


Figure 4-20 SDC2 and SDC4 schematic

4.9.5 Configurable GPIO

Configurable GPIO ports and MPM support

There are 180 GPIO ports – 146 of them are normal GPIOs, and configuration options are shown here and 34 of them are called extended GPIOs (eGPIO).

Software assigns functions to the GPIOs, and their configurations are set accordingly.

If MPM is enabled by software to reduce power consumption, only selected GPIO pins can be used to wake up the QRB5165 device. See the *QRB5165 Data Sheet* (80-PV086-1) for wake-up pin information.

If MPM is used, treat external pulls carefully. During MPM mode, all digital pads are held by a keeper circuit. If the MPM GPIO is driven by an external chip during MPM mode, external opposite-direction pull causes DC current flows (highly undesirable during sleep mode). If the MPM GPIO is connected to an open-drain circuit, an external pull is required for keeping the GPIO in the intended state.

The PCB layout must avoid crosstalk on the wake-up GPIOs; coupling can cause an unintentional trigger.

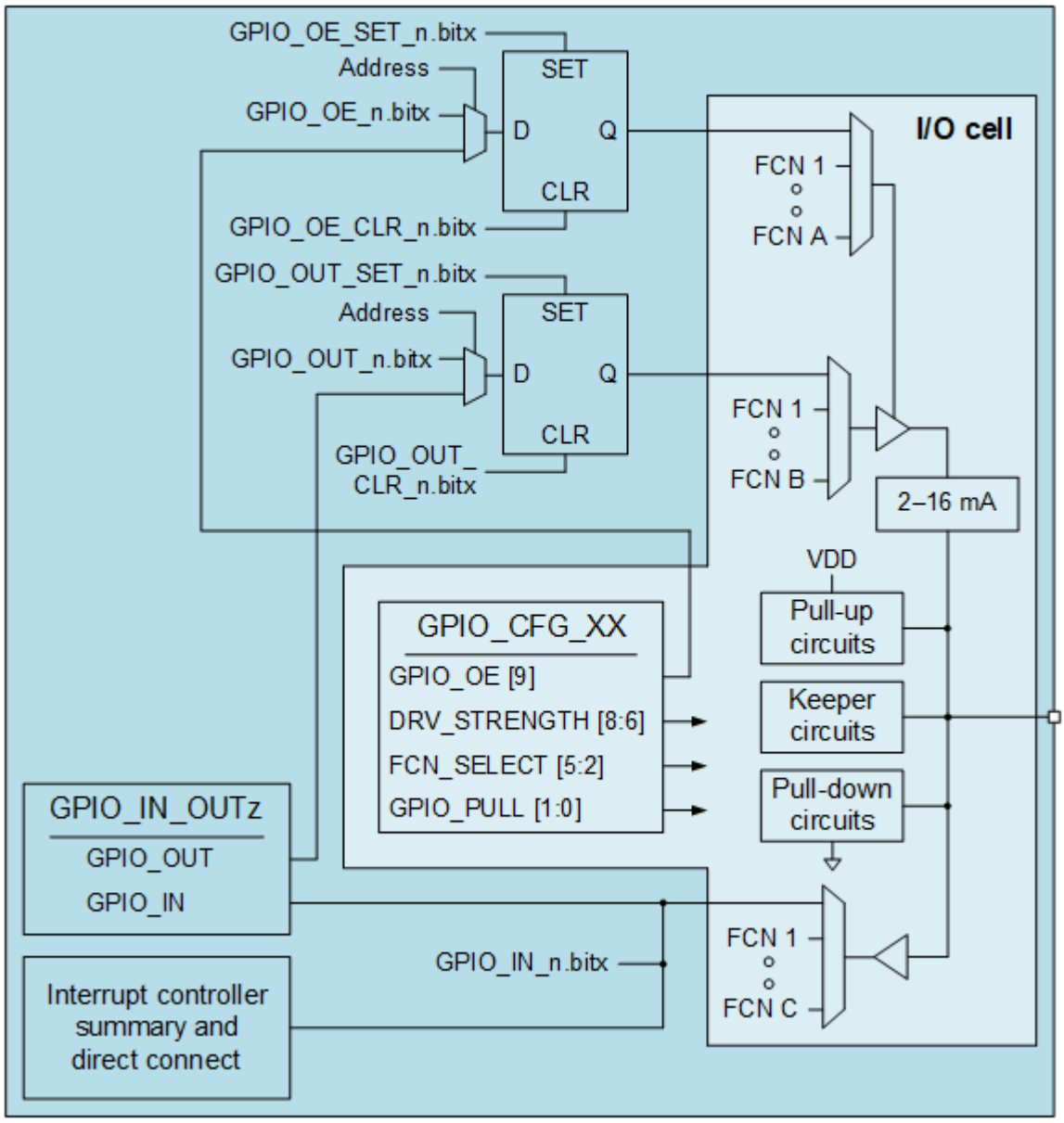


Figure 4-21 Configurable GPIO ports and MPM support

GPIO initialization

To avoid GPIO problems, a specific supply sequence is required when powering up the QRB5165 device.

- The core supply (VDD_CX) must turn on and reach 90% of its programmed value before any of the pad supplies (VDD_PX) are enabled. Without these sequence and timing relationships the GPIO pads might come up in undefined states.
- The PM8250 and PM8150L/PM8150A ensure the proper supply sequence and timing.
- Recommend peripheral devices connected to GPIOs not driving the signals until RESIN_N de-asserted.

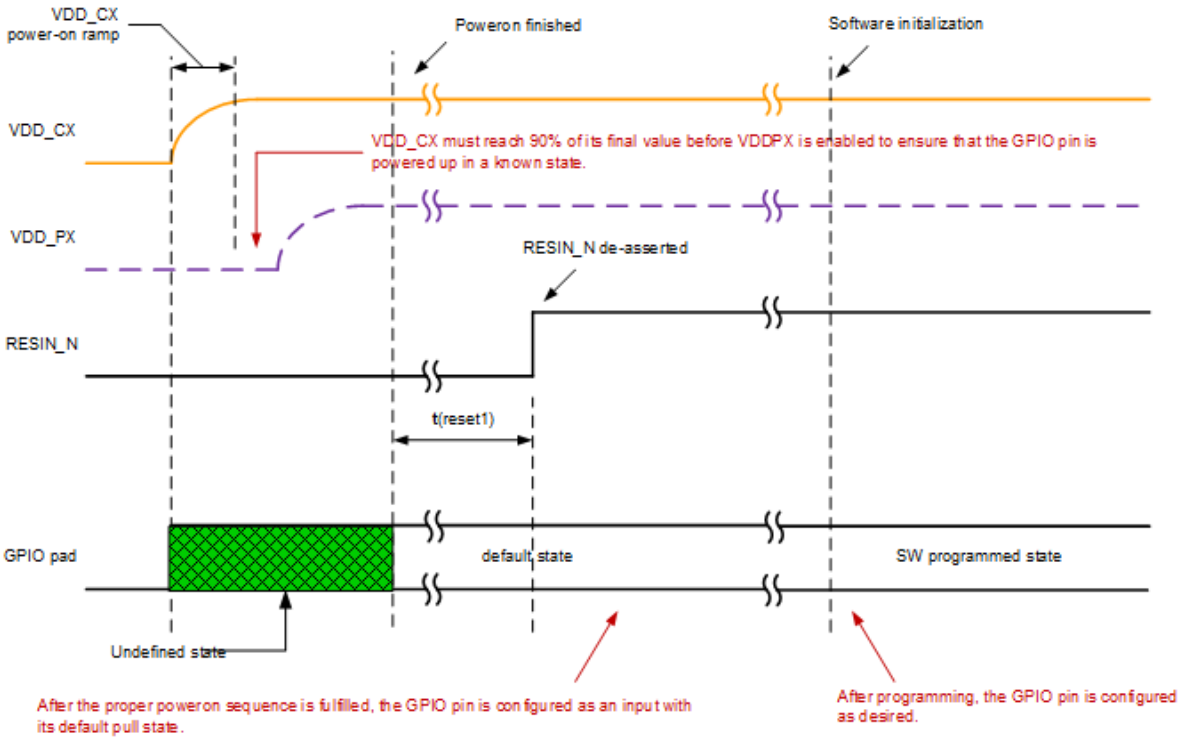


Figure 4-22 GPIO initialization

eGPIO ports

There are 34 extended GPIO (eGPIO) ports:

- eGPIO allows SSC or LPASS I/Os to be reused as regular GPIOs, if intended.
- 20 of them are configured as SSC GPIOs by default.
- 14 of them are configured as LPASS GPIOs by default.

Two new bits are added to each GPIO_CFG register for eGPIO function.

- EGPIO_PRESENT is the read-only bit which shows if an eGPIO feature is available on a particular GPIO. This bit will be 1 only for GPIO 146 to GPIO 179 and will be 0 for GPIO 0 to GPIO 145 in QRB5165.
- EGPIO_ENABLE is the read/write bit and only effective if EGPIO_PRESENT is 1. Once it is set, the eGPIO is controlled from the chip TLMM block instead of SSC or WCSS.

See the *QRB5165 Pin Assignment and GPIO Configuration Spreadsheet* (80-PV086-1A) and *QRB5165 Hardware Register Description* (80-PV086-2X) for more information on eGPIOs.

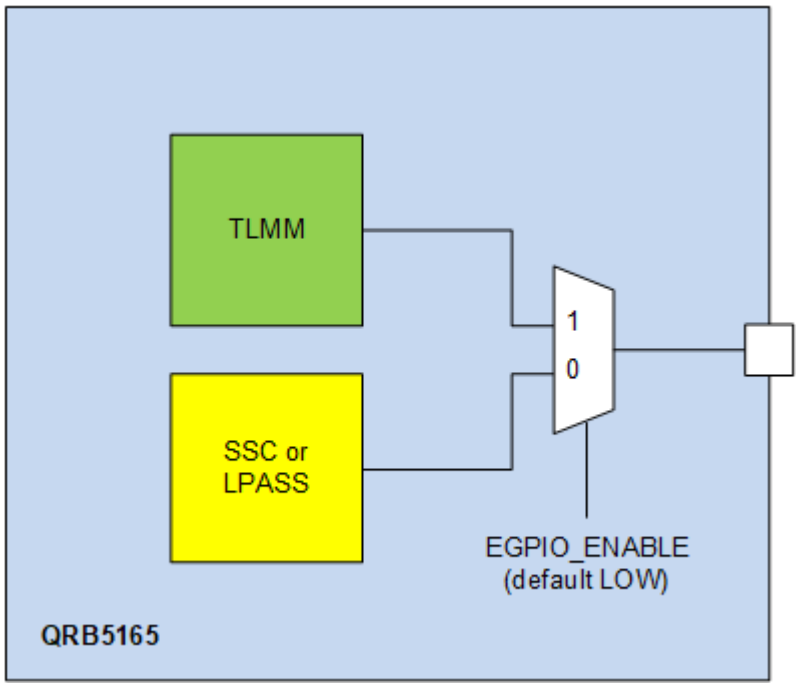


Figure 4-23 eGPIO ports

5 QRB5165 interfaces

5.1 QRB5165 interface introduction

The QRB5165 interfaces with other chipset ICs as follows:

Table 5-1 Chipset interfaces

Chipset	Interface
4G/5G Modem IC	PCIe2 and UART(s)
QCA639x (WLAN/ Bluetooth IC)	PCIe0, SLIMbus, and UART
QCA643x + QCA642x WiGig	PCIe1
Power management	■ SPMI for PM8250, PM8150L/PM8150A, PM8150B, PMK8002, PMX55, and PM8009 ■ I^2C for SMB1355 and SMB1390
Audio codec	SWR for WCD9380
Smart speaker amplifier	SWR for WSA881x

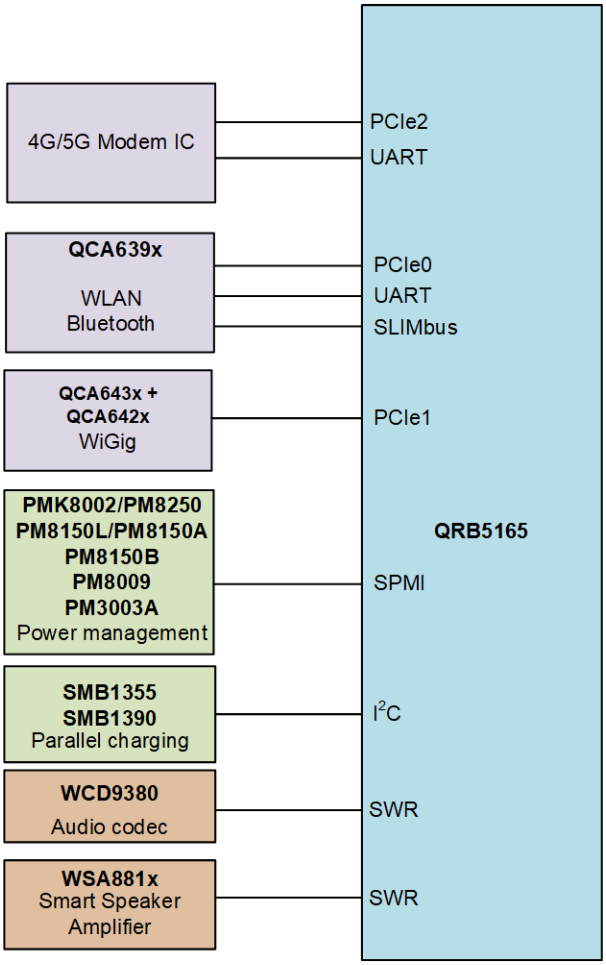


Figure 5-1 QRB5165 interface with other chipset IC

5.2 QRB5165 PMIC interfaces

The following figure shows different PMIC chip interfaces with each other.

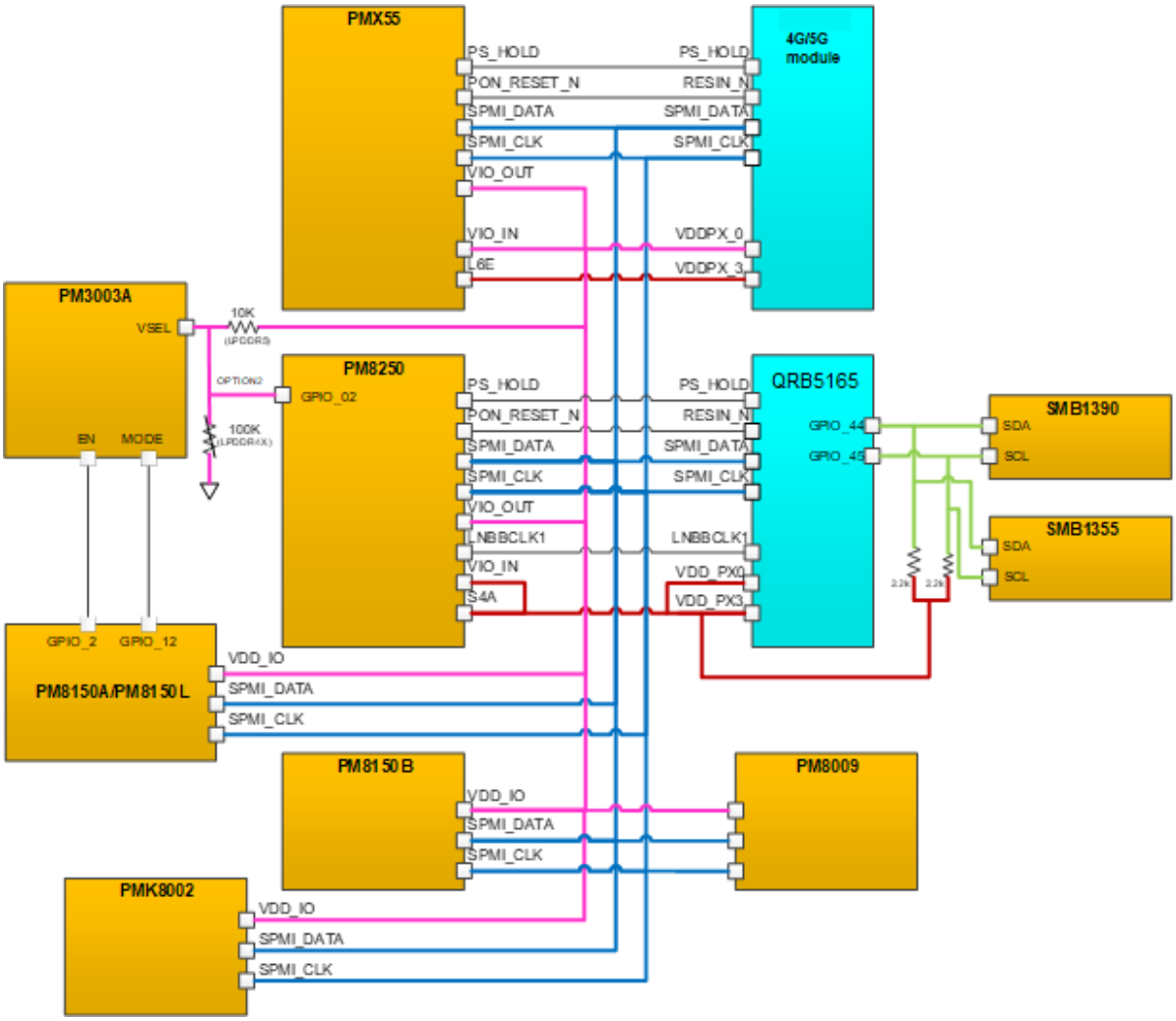


Figure 5-2 PMIC interfaces

NOTE Back-up interfaces are not shown in this figure. See the *Design Package, QRD8250, Smart Phone, Electrical Hardware, PCB Schematic & Layout* (DP10-VL764-100A) for details.

5.3 QRB5165 wireless connectivity interfaces

The following figure shows the wireless connectivity interface of QRB5165.

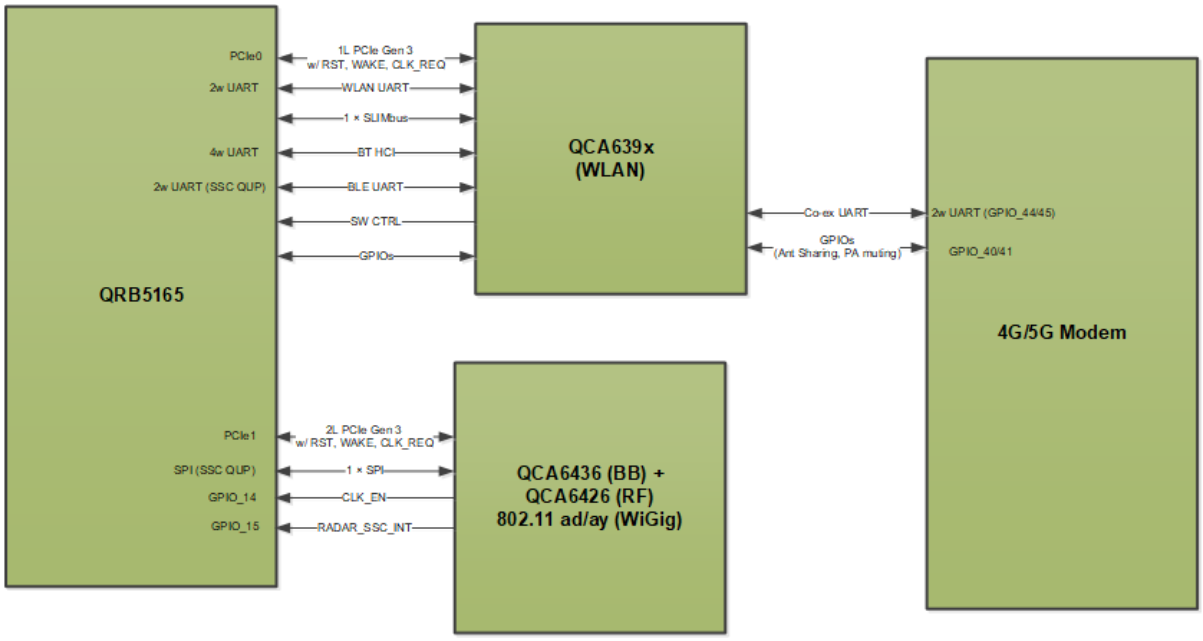


Figure 5-3 QRB5165 wireless connectivity interface

5.3.1 Near field communications (NFC)

QTI platforms use NXP SN100U parts for NFC.

Table 5-2 NFC parts

SN100U	QRB5165
NFC_I2C_SDA	GPIO_4
NFC_I2C_SC	GPIO_5
NFC_IRQ	GPIO_111
NFC_EN	GPIO_6
NFC_DWL_REQ	GPIO_110
NFC_CLK_REQ	GPIO_7
	PM8250
19.2 MHZ_CLK (NFC_XTAL1)	LN_BB_CLK3
eSE	
SN100U eSE	**QRB5165**
ESE_SPI_MISO	GPIO_28
ESE_SPI_MOSI	GPIO_29

Table 5-2 NFC parts (cont.)

SN100U	QRB5165
ESE_SPI_CLK	GPIO_30
ESE_SPI_CS_N	GPIO_31

5.4 QRB5165 audio interfaces

For details on QRB5165 LPASS, see the *SM8250/SXR2130P/QRB5165/QRB5165N/QCS8250/QCS8245 Audio Codec Hardware Overview Design Guidelines/Training Slides* (80-PL546-5C).

For design details, see the *Qualcomm Aqstic WCD9380/WCD9385 Audio Codec Hardware Design Guidelines/Training Slides* (80-PL335-5).

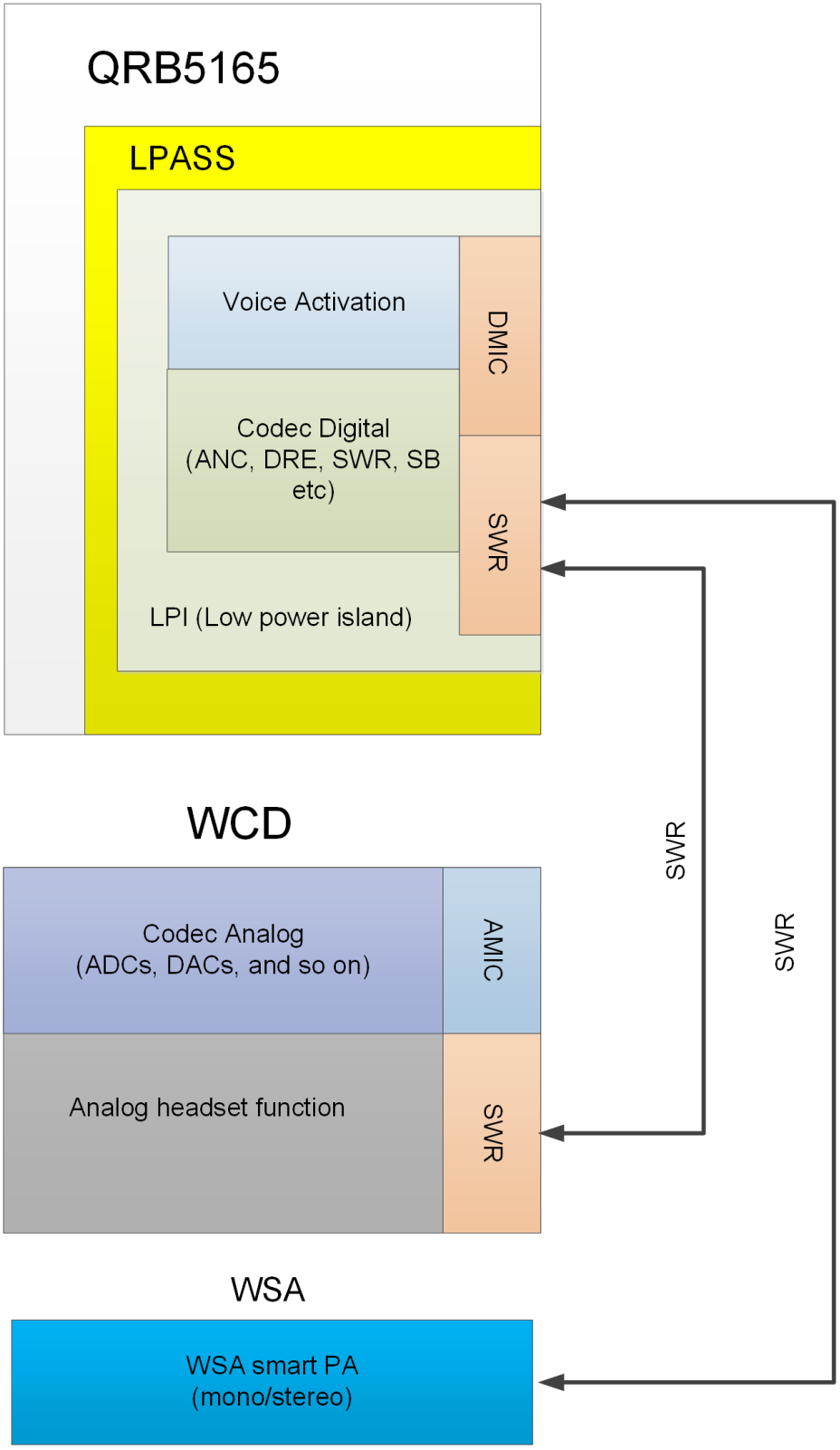


Figure 5-4 Audio interfaces

6 Power management

6.1 QRB5165 power management ICs

The QRB5165 chipset has the following PMICs:

- SPMI for PM8250, PM8150L/PM8150A, PM8150B, PMK8002, PMX55, and PM8009
- I^2C for SMB1355 and SMB1390

The following are the details listed for each PMIC:

- Even distribution between two core PMICs (PM8250 and PM8150L/PM8150A)
- Improved PMIC partitioning for better routing efficiency, less noise, and better thermal
  - Display power and flash located at the top
  - Charger located at the bottom next to the Type-C connector
- PM8250 is optimized for QRB5165 chipset
  - The only change of PM8250 from PM8150 is the internal OTP memory to support power grid and power-on sequence required for QRB5165 chipset.
- Only one of either PM8150L or PM8150A is required in the system.
  - PM8150L is dedicated for LCD display power supply in addition to serving as a core PMIC.
  - PM8150A is dedicated for AMOLED display power supply in addition to serving as a core PMIC.
- PM8009: for QCA639x and camera
  - 2 HF-SMPS
  - LDOs
- PM3003A-5: Buck regulator for a DDR rail

6.2 PM8150B and parallel charging solution

The PM8150B provides best-in-class charging performance while integrating many related functions.

- 12 V charger (28 V over voltage protection (OVP)) supports up to 10 A multipath charge current
- Qualcomm® Quick Charge™ 2, Quick Charge 3, Quick Charge 4 + USB PD support
- Type-C detection and 20 V CC/SBU pin protection

- 3 A USB host mode support
- Improved hardware fuel gauge
- Qnovo pulse engine support

The SMB1390 features are listed below:

- 2:1 switched capacitor converter for high current, low profile Li+ battery charging and voltage regulation
- Master/slave mode to minimize input current ripple and support 10 A battery charging with low dissipation power
- PM8150B can pair with SMB1355 and/or SMB1390 for parallel charging

Hardware configuration	Quick Charge 4 (PD PPS)	Quick Charge 3	Quick Charge 2/PD [a]/DCP	Comments
PM8150B standalone	PM8150B			
PM8150B + SMB1355	PM8150B + SMB1355			
PM8150B + SMB1390	PM8150B + SMB1390 [b]	PM8150B + SMB1390 [b]	PM8150B	For up to 6 A charging
PM8150B + SMB1355 + SMB1390	PM8150B + SMB1390 [b]	PM8150B + SMB1390 [b]	PM8150B + SMB1355	
PM8150B + 2x SMB1390	PM8150B + 2x SMB1390 [b]	PM8150B + 2x SMB1390 [b]	PM8150B	For 8 A charging with a 5 A cable

[a] The Vanilla PD that offers 5 V, 9 V, 12 V, and so on.
[b] Switch to PM8150B during taper charging for end-of-charge and recharge.

7 Battery charging

7.1 SMB1390 battery charging overview

The SMB1390 is a single-input, programmable single-cell Lithium-ion/Lithium-polymer battery charger. SMB1390 is a divide-by-two charge pump, providing the highest efficiency for charging high-capacity cells, up to 6 A fast charge capability per SMB1390. The device provides +28 V DC withstand, wide operating input range of 3.6 to 12 V, and allows for the elimination of inductors common to buck regulator solutions.

7.2 SMB1390 battery charging features

The SMB1390 primary features are:

- +2.5 V to +5 V operating output voltage range (input voltage = 2 × output voltage + delta)
- +28 V withstand input voltage with 2.5 V on the output
- High efficiency with a constant 2:1 conversion ratio
- Programmable switching frequency
  - 100 kHz to 2 MHz in normal operation mode
  - 12.5 kHz to 100 kHz in low current mode
- Up to 6 A output current per device, expandable with dual implementation
- Fully integrated, low R_{DSON} FETs switching
- Dual charging (two SMB1390s charging) operation with clock sync to reduce overall current ripple
- Protection features:
  - Programmable 0.5 A to 3.2 A input overcurrent protection (OVP), 100 mA per step
  - Programmable output overvoltage protection
  - Soft start
  - Input overvoltage lockout
  - Input undervoltage lockout
  - Thermal shutdown
  - Reverse current protection
  - Die temperature sensing pin

- 400 µA typical quiescent current standby
- 1 µA typical quiescent current shutdown

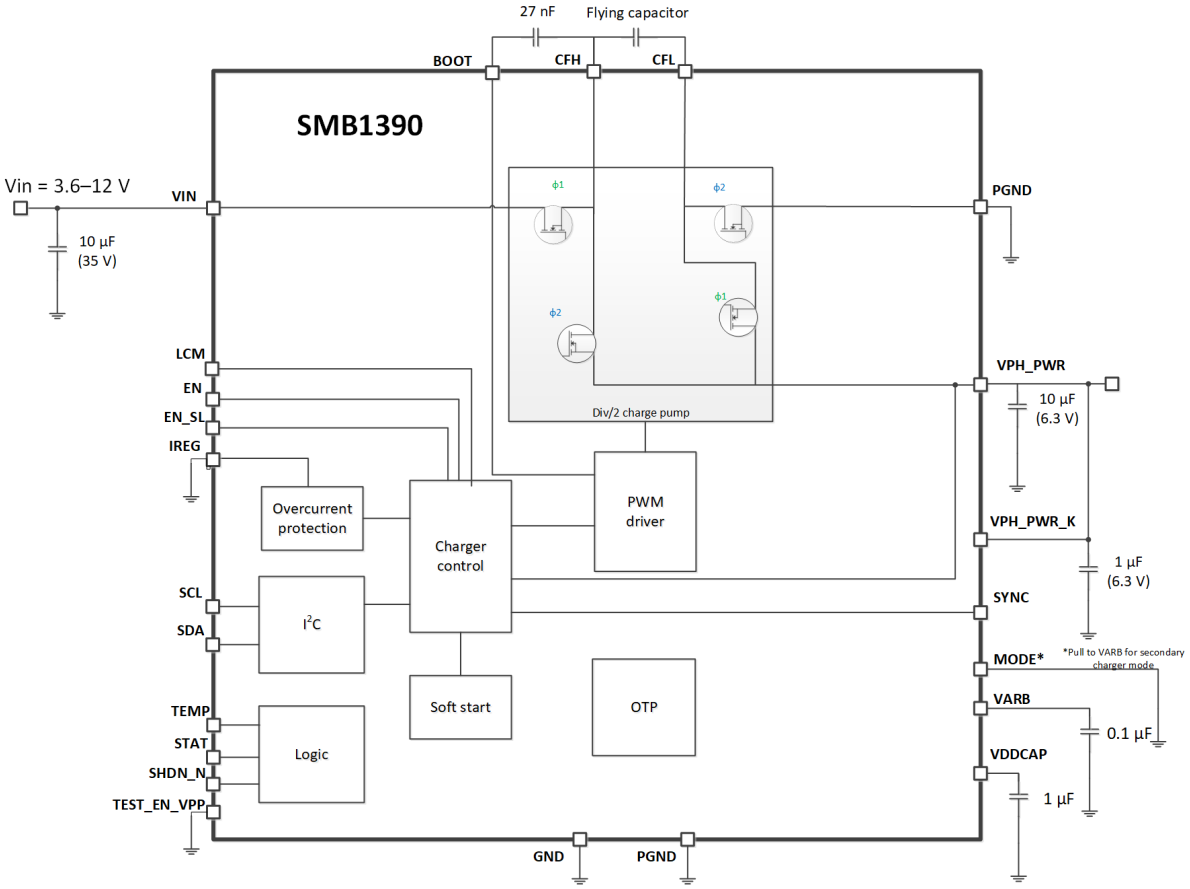


Figure 7-1 SMB1390 block diagram

8 Design guidelines

8.1 PCB design considerations

See the following checklists through the design process and before the PCB tapeout.

QRB5165 Schematic Checklist

The schematic checklist is an accumulation of notes from QTI internal reference design and testing. The QTI CE team will use this checklist to verify each customer-submitted schematic design. Consider each of these items when doing your individual product designs. To view the schematic checklist, click *QRB5165 Schematic Review Checklist* (80-PV086-111).

QRB5165 Layout Checklist

The layout review checklist is an accumulation of notes from QTI internal reference design and testing. The QTI CE team will use this checklist to validate each customer-submitted layout design. Consider each of these items when doing your individual product designs. To view the layout review checklist, click *QRB5165 Layout Review Checklist* (80-PV086-112).

8.1.1 Critical component placement

The QRB5165 critical component placement is listed below.

- The following components should be placed close to their respective QRB5165 pins.
- In most cases, these signals should route to their components on layer 1 or layer 2 of the PCB.
- ZQ and EBIx_CAL resistors tie to the VDDQ supply for LPDDR5 PoP memories.
- All the REXT resistors should be placed as close to the QRB5165 ball as possible, and the REXT routing should be short and isolated.

Table 8-1 Critical component placement

QRB5165 pin name	Component requirement	Pull-up/pull-down/series connection
EBI02_CAL	240 Ω ± 1%	Pull-up – VDDQ
EBI13_CAL	240 Ω ± 1%	Pull-up – VDDQ
QREFS_CXO_REXT	3 kΩ ± 1%	Pull-down
REFGEN_REXT0	100 Ω ± 1%	Pull-down
REFGEN_REXT1	100 Ω ± 1%	Pull-down

Table 8-1 Critical component placement (cont.)

QRB5165 pin name	Component requirement	Pull-up/pull-down/series connection
ZQ_A (LPDDR5)	240 Ω ± 1%	Pull-up – VDDQ
ZQ_D (LPDDR5)	240 Ω ± 1%	Pull-up – VDDQ

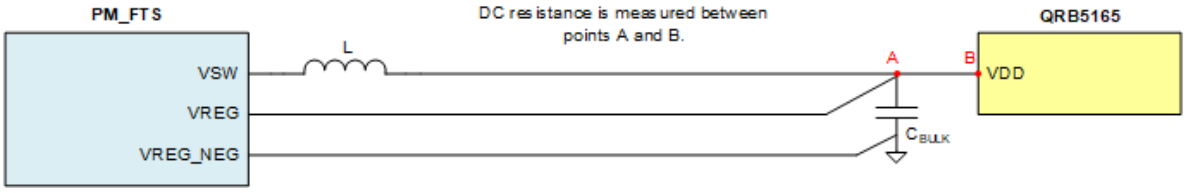
8.1.2 QRB5165 PDN specifications

See *QRB5165 Data Sheet* (80-PV086-1) for PDN specifications.

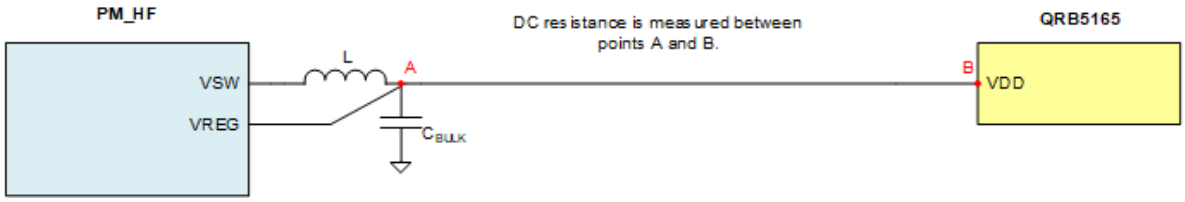
8.1.3 DCR measurement for PDN rails

The FTS bucks are high-current regulators that use remote sense to help achieve low DCR PDN specifications on the PCB.

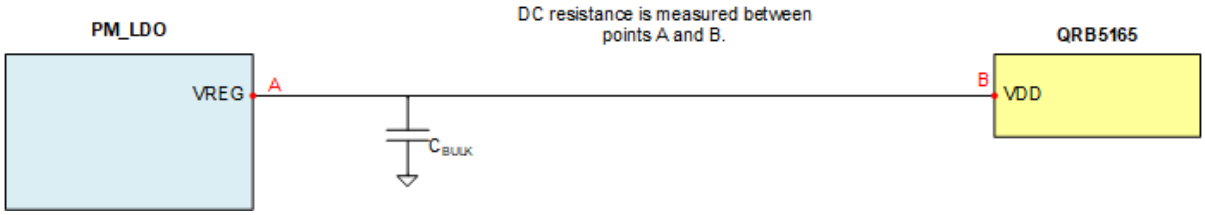
The bulk capacitor must be placed close to the load, and a corresponding sense line must connect to the domain at the bulk capacitor.



The HF buck regulators use a local sense setup, which means the bulk capacitor for the buck remains adjacent to its inductor.



The LDOs do not have any external sense nodes. The LDO DC resistance should be measured between the LDO pin on the PMIC and the load pins on the SDM.



8.1.4 Unused pin termination

The following table lists the unused pin termination in various interfaces.

NOTE If a PHY is not used, then proper software change needs to be made to keep the PHY in the off state.

Table 8-2 Unused pin termination

Interface	Signal	Unused pin state
UFS0/UFS1		
UFS0	UFS0_RX0_M	Float
	UFS0_RX0_P	Float
	UFS0_TX0_M	Float
	UFS0_TX0_P	Float
	UFS0_RX1_M	Float
	UFS0_RX1_P	Float
	UFS0_TX1_M	Float
	UFS0_TX1_P	Float
	UFS0_REFCLK	Float
	UFS0_RESET	Float
	VDDA_UFS0_1P2	Connect to power supply
	VDDA_UFS0_CORE	Connect to power supply
UFS1	UFS1_REFCLK	Float
	UFS1_RX0_M	Float
	UFS1_RX0_P	Float
	UFS1_TX0_M	Float
	UFS1_TX0_P	Float
	VDDA_UFS1_1P2	Connect to power supply
	VDDA_UFS1_CORE	Connect to power supply
SDC2		
SDC2	SDC2_CLK	Float
	SDC2_CMD	Float
	SDC2_DATA0	Float
	SDC2_DATA1	Float
	SDC2_DATA2	Float
	SDC2_DATA3	Float
	VDDPX_VBIAS_SDC	GND
	VDDPX_2	GND
MIPI DSI0/DSI1		
MIPI DSI0	DSI0_A0_LN0_P	Float
	DSI0_B0_LN0_N	Float
	DSI0_C0_LN1_P	Float
	DSI0_A1_LN1_N	Float
	DSI0_B1_CLK_P	Float
	DSI0_C1_CLK_N	Float

Table 8-2 Unused pin termination (cont.)

Interface	Signal	Unused pin state
	DSI0_A2_LN2_P	Float
	DSI0_B2_LN2_N	Float
	DSI0_C2_LN3_P	Float
	DSI0_NC_LN3_N	Float
MIPI DSI1	DSI1_A0_LN0_P	Float
	DSI1_B0_LN0_N	Float
	DSI1_C0_LN1_P	Float
	DSI1_A1_LN1_N	Float
	DSI1_B1_CLK_P	Float
	DSI1_C1_CLK_N	Float
	DSI1_A2_LN2_P	Float
	DSI1_B2_LN2_N	Float
	DSI1_C2_LN3_P	Float
	DSI1_NC_LN3_N	Float
	VDDA_DSI_1P2	Connect to power supply
	VDDA_DSI_0P9	Connect to power supply
	VDDA_DSI_PLL_0P9	Connect to power supply
USB and DisplayPort		
USB0_SS_DP	USB0_SS_TX0_M	Float
	USB0_SS_TX0_P	Float
	USB0_SS_RX0_M	Float
	USB0_SS_RX0_P	Float
	USB0_SS_TX1_M	Float
	USB0_SS_TX1_P	Float
	USB0_SS_RX1_M	Float
	USB0_SS_RX1_P	Float
	VDDA_USB0_SS_DP_1P2	Connect to power supply
	VDDA_USB0_SS_DP_CORE	Connect to power supply
USB0_HS	USB0_HS_DM	Float
	USB0_HS_DP	Float
	VDDA_USB_HS_1P8	Connect to power supply
	VDDA_USB_HS_3P1	Connect to power supply
	VDD_USB_HS_CORE	Connect to power supply
USB1_SS	USB1_SS_TX_M	Float
	USB1_SS_TX_P	Float
	USB1_SS_RX_M	Float

Table 8-2 Unused pin termination (cont.)

Interface	Signal	Unused pin state
	USB1_SS_RX_P	Float
	VDDA_USB1_SS_1P2	Connect to power supply
	VDDA_USB1_SS_CORE	Connect to power supply
USB1_HS	USB1_HS_DM	Float
	USB1_HS_DP	Float
	VDDA_USB_HS_1P8	Connect to power supply
	VDDA_USB_HS_3P1	Connect to power supply
	VDD_USB_HS_CORE	Connect to power supply
DisplayPort	DP_AUX_P	Float
	DP_AUX_N	Float
MIPI CSI0	CSI0_NC_CLK_P	Float
	CSI0_A0_CLK_N	Float
	CSI0_B0_LN0_P	Float
	CSI0_C0_LN0_N	Float
	CSI0_A1_LN1_P	Float
	CSI0_B1_LN1_N	Float
	CSI0_C1_LN2_P	Float
	CSI0_A2_LN2_N	Float
	CSI0_B2_LN3_P	Float
	CSI0_C2_LN3_N	Float
MIPI CSI1	CSI1_NC_CLK_P	Float
	CSI1_A0_CLK_N	Float
	CSI1_B0_LN0_P	Float
	CSI1_C0_LN0_N	Float
	CSI1_A1_LN1_P	Float
	CSI1_B1_LN1_N	Float
	CSI1_C1_LN2_P	Float
	CSI1_A2_LN2_N	Float
	CSI1_B2_LN3_P	Float
	CSI1_C2_LN3_N	Float
MIPI CSI2	CSI2_NC_CLK_P	Float
	CSI2_A0_CLK_N	Float
	CSI2_B0_LN0_P	Float
	CSI2_C0_LN0_N	Float
	CSI2_A1_LN1_P	Float
	CSI2_B1_LN1_N	Float

Table 8-2 Unused pin termination (cont.)

Interface	Signal	Unused pin state
	CSI2_C1_LN2_P	Float
	CSI2_A2_LN2_N	Float
	CSI2_B2_LN3_P	Float
	CSI2_C2_LN3_N	Float
MIPI CSI3	CSI3_NC_CLK_P	Float
	CSI3_A0_CLK_N	Float
	CSI3_B0_LN0_P	Float
	CSI3_C0_LN0_N	Float
	CSI3_A1_LN1_P	Float
	CSI3_B1_LN1_N	Float
	CSI3_C1_LN2_P	Float
	CSI3_A2_LN2_N	Float
	CSI3_B2_LN3_P	Float
	CSI3_C2_LN3_N	Float
MIPI CSI4	CSI4_NC_CLK_P	Float
	CSI4_A0_CLK_N	Float
	CSI4_B0_LN0_P	Float
	CSI4_C0_LN0_N	Float
	CSI4_A1_LN1_P	Float
	CSI4_B1_LN1_N	Float
	CSI4_C1_LN2_P	Float
	CSI4_A2_LN2_N	Float
	CSI4_B2_LN3_P	Float
	CSI2_C2_LN3_N	Float
MIPI CSI5	CSI5_NC_CLK_P	Float
	CSI5_A0_CLK_N	Float
	CSI5_B0_LN0_P	Float
	CSI5_C0_LN0_N	Float
	CSI5_A1_LN1_P	Float
	CSI5_B1_LN1_N	Float
	CSI5_C1_LN2_P	Float
	CSI5_A2_LN2_N	Float
	CSI5_B2_LN3_P	Float
	CSI5_C2_LN3_N	Float
CSI power	VDD_A_CSI0_0P9	Connect to power supply
	VDD_A_CSI012_1P2	Connect to power supply

Table 8-2 Unused pin termination (cont.)

Interface	Signal	Unused pin state
	VDD_A_CSI1_2_0P9	Connect to power supply
	VDD_A_CSI3_0P9	Connect to power supply
	VDD_A_CSI345_1P2	Connect to power supply
	VDD_A_CSI4_5_0P9	Connect to power supply
PCIE0	PCIE0_REFCLK_M	Float
	PCIE0_REFCLK_P	Float
	PCIE0_TX_M	Float
	PCIE0_TX_P	Float
	PCIE0_RX_M	Float
	PCIE0_RX_P	Float
	VDDA_PCIE0_PLL_1P2	Connect to power supply
	VDDA_PCIE0_CORE	Connect to power supply
PCIE1	PCIE1_REFCLK_M	Float
	PCIE1_REFCLK_P	Float
	PCIE1_TX0/1_M	Float
	PCIE1_TX0/1_P	Float
	PCIE1_RX0/1_M	Float
	PCIE1_RX0/1_P	Float
	VDDA_PCIE1_PLL_1P2	Connect to power supply
	VDDA_PCIE1_CORE	Connect to power supply
PCIE2	PCIE2_REFCLK_M	Float
	PCIE2_REFCLK_P	Float
	PCIE2_TX0/1_M	Float
	PCIE2_TX0/1_P	Float
	PCIE2_RX0/1_M	Float
	PCIE2_RX0/1_P	Float
	VDDA_PCIE2_PLL_1P2	Connect to power supply
	VDDA_PCIE2_CORE	Connect to power supply

8.2 PCB layout guidelines

8.2.1 PCB floor plan recommendation (C-shape)

The QRB5165 chip can be placed close to the top of the board with PM8150L/PM8150A on the north side and PM8250 with WWAN on the south side, which fully aligns with the package pin map. The PM8250 and PM8150L/PM8150A should be placed on the opposite side of QRB5165.

The camera connectors can be placed on the north side of the PCB and display connectors can be placed on the south side of the PCB. The decoupling capacitors are placed opposite the QRB5165 region. The UFS is placed on the top side, south to the QRB5165. The image below shows the measurements in mm.

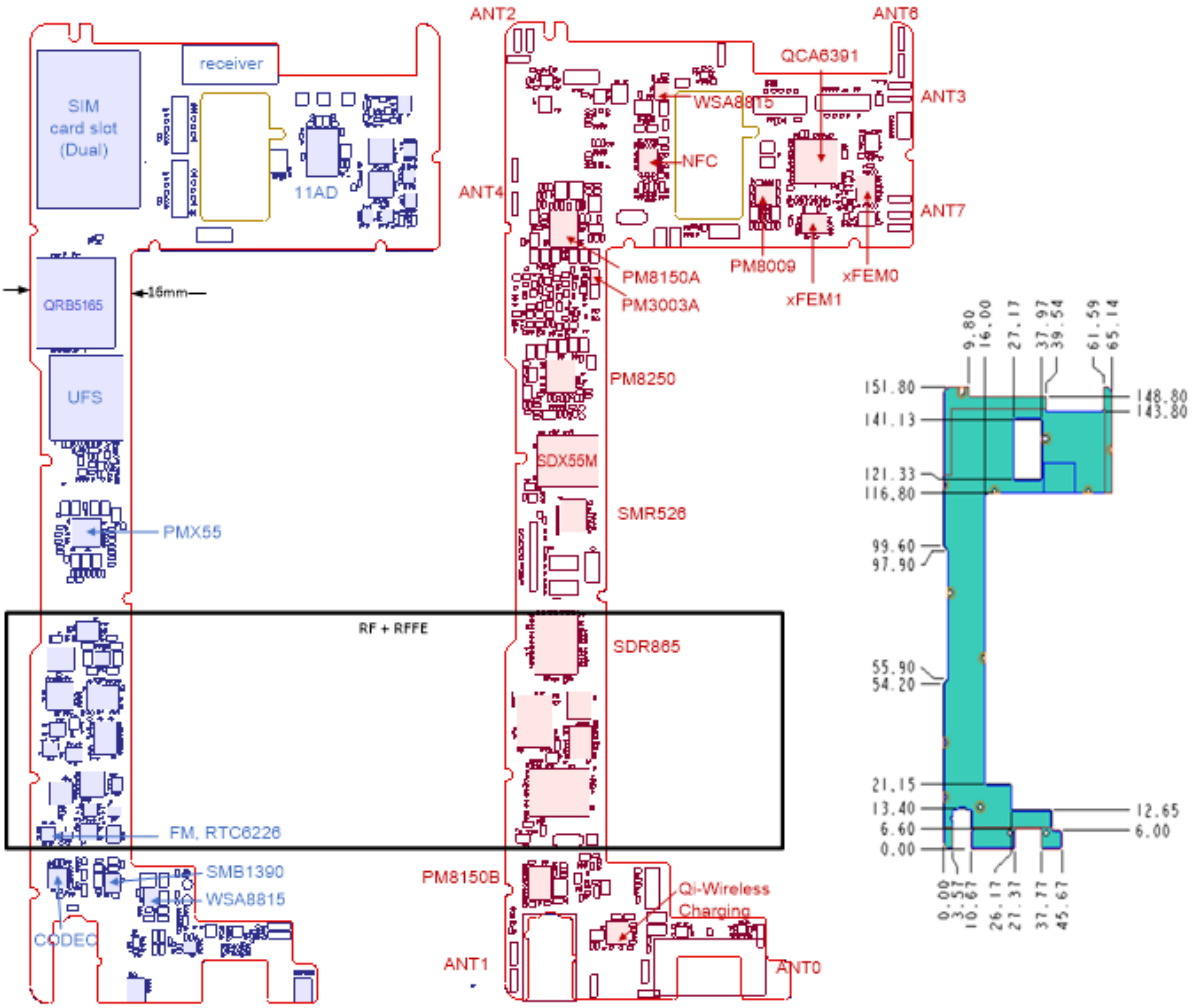


Figure 8-1 PCB floor plan recommendation (C-shape)

8.2.2 PCB floor plan recommendation (H-shape)

The QRB5165 chip can be placed on the middle of the board, with PM8150L/PM8150A on the north east side and PM8250 on the south east side, which fully aligns with the package pin map. The PM8250 and PM8150L/PM8150A should be placed on the opposite side of QRB5165.

The camera connectors can be placed on the north side of the PCB and display connectors can be placed on the south side of the PCB. The decoupling capacitors are placed opposite the SDM region. The UFS is placed on the east side to the SDM.



Figure 8-2 PCB floor plan recommendation (H-shape)

8.2.3 Landside capacitor (LSC) keepout area routing rules

The LSC keepout area must be left without any exposed pad, trace, or via

- No exposed pad under LSC keepout areas
- No exposed trace under LSC keepout areas
- No exposed vias under LSC keepout areas

Any pad/trace/vias under the LSC keepout area must be covered by the solder mask. The copper clearance under the LSC keepout area is the recommended rule. For more information, see the *SMT Assembly Guidelines* (SM80-P0982-1).

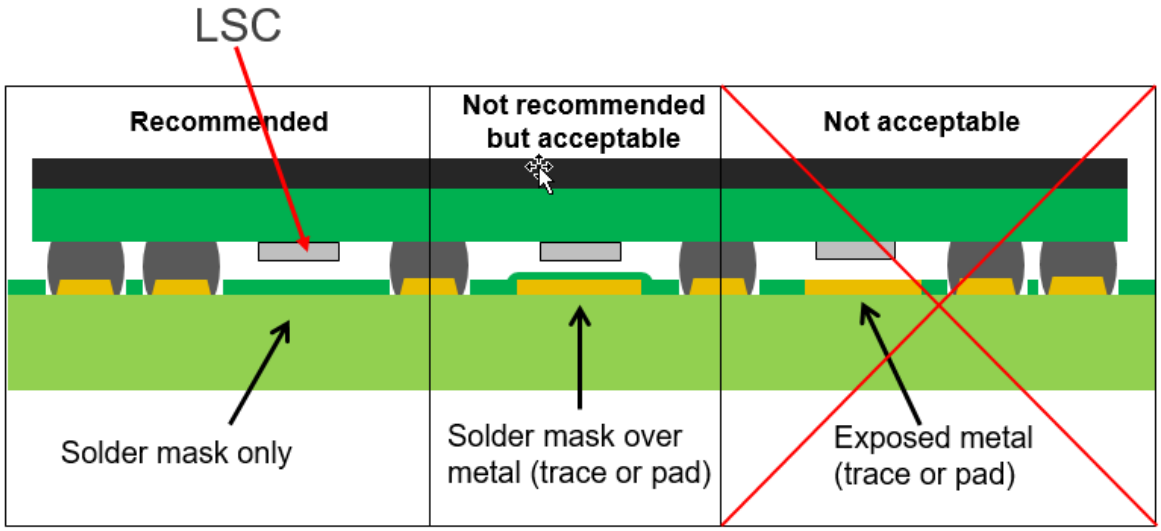


Figure 8-3 LSC keepout area routing rules

8.2.4 CXO and SLEEP_CLK routing

All the CLK vias should be completely surrounded as they transition between layers. A C-shaped ring of vias should be standard for all clock and RF line transitions.

The QRB5165 device has two main clock inputs: CXO and SLEEP_CLK. Route each of these clocks from the PM8250 to the QRB5165, while referencing a solid GND plane.

- These signals should be routed away from sensitive nets.
- The PM8250 clock driver output impedance can be controlled to match the line impedance and mitigate reflections.
- Both CXO and SLEEP_CLK should immediately via down to inner layers when escaping the QRB5165 to remain isolated from the rest of the BGA fan-out.

SLEEP_CLK signal routing guidelines

The SLEEP_CLK signal routing guidelines are listed below.

- The SLEEP_CLK buffer is redesigned to have a much slower edge rate; therefore, it is more favorable to multidestination routing of the clocks.
- Either star-routing or daisy-chain routing is acceptable.
  - If star-routing is used, ensure that the two-branch length difference is not too long (shorter than 30 mm is recommended).
  - If daisy-chain routing is used, ensure that the stub is as short as possible (shorter than 7 mm is recommended).

8.2.5 SS-USB and HS-USB with PHY layout guidelines

The following are the SS-USB and HS-USB guidelines:

- External components must be located near the USB connector.
- There are relatively fast edge rates, so they must be routed away from sensitive circuits and signals (RF, audio, and 38.4 MHz XO).
- If a USB connector is used as a charger input:
  - The USB_VBUS node must be routed to the PMIC device using extremely wide traces or sub-planes.
  - For fetailed recommendations see the *PM8150/PM8250 Power Management IC Design Guidelines/Training Slides* (80-PD995-5).
- Even if the USB connector is not used for charging, USB_VBUS can be used as the power bus for the USB. The trace width must be sized depending on the length of VBUS and the expected current.
  - USB peripheral currents can be up to 900 mA for SS and 500 mA for HS.

Additional SS-USB guidelines

- Maintain good isolation between the USB connector and RF antennas (especially 2.4 GHz).
- Route the RF signals operating at a 2.4 GHz frequency with the highest isolation possible from USB0_SS_TX/RX and USB1_SS_TX/RX traces.

- USB SS Tx AC coupling can be anywhere along the line, but better be placed close to source or the ESD/connector side to keep good signal integrity (SI) of main route on PCB.
- Route differential pairs in the inner layers with a solid GND reference to have good impedance control and to minimize discontinuities.
- Keep isolation between the Tx pair, Rx pair, and DP/DM to avoid crosstalk.
- Keep the 1 kΩ resistor stubs on DP/DM lines as short as possible.
- If core vias are used, use no more than two core vias per signal line to limit stubs.
- If sending USB over a flex PCB, be extra careful to meet the entire channel insertion loss (IL) budget. A poor flex will violate USB IL specifications and may force the need for a re-driver circuit to be added to the eBOM.

8.2.6 HS-USB DP/DM line lengths

The following figure shows the HS-USB DP/DM line lengths.

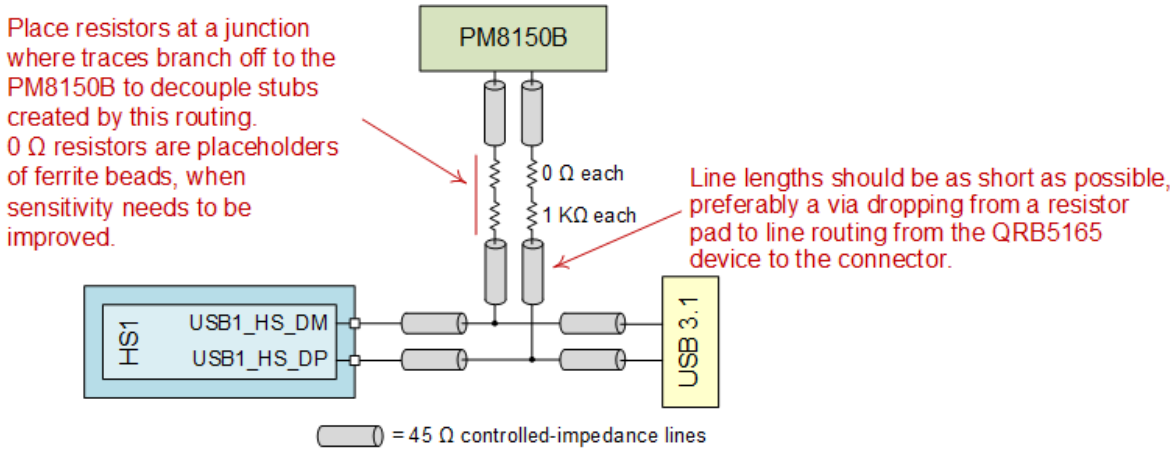


Figure 8-4 HS-USB DP/DM line lengths

8.2.7 USB 3.1/DisplayPort routing constraints

The following table captures the routing constraints of USB 3.1 and DisplayPort.

Table 8-3 USB 3.1/DisplayPort routing constraints

Metrics			Guidance	
Data rate			10 Gbps (Gen 2)	5 Gbps (Gen 1)
Impedance [a]	Differential	Field route	70 to 100 Ω	
Channel insertion loss budget			-7 dB at 5 GHz	-5.5 dB at 2.5 GHz
Length matching	USB SS or DisplayPort data lane intralane match		5 ps (700 µm)	
	USB SS or DisplayPort data lane interlane match		70 ps (10 mm)	
	DP_AUX_P/DP_AUX_M intralane match		50 ps (7 mm)	
Spacing	Rx-to-Tx	Field route	4 × line width	3 × line width

Table 8-3 USB 3.1/DisplayPort routing constraints (cont.)

Metrics			Guidance	
	Space to other signals		4 × line width	
Component	AC capacitor		75 to 265 nF	75 to 200 nF

[a] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.8 USB 2.0 routing constraints

The following table captures the routing constraints of USB 2.0 and DisplayPort.

Table 8-4 USB 2.0 routing guidelines

Metrics			Guidance
Data rate			480 Mbps
Impedance	Differential	Field route	75 – 105 Ω
Length match	Intralane match		2 mm
Maximum PCB trace length			250 mm
Spacing	To other signals	Field route	3 × line width

8.2.9 MIPI_CSI D-PHY routing constraints (up to 2.5 Gbps)

To interpret the following table, use the 500 Mbps/lane as an example. If cable insertion loss is -0.5 dB or better, the trace length needs to be ≤ 260 mm. If cable insertion loss is between -0.5 dB and -1.0 dB, the trace length needs to be ≤ 190 mm. If cable insertion loss is worse than -1.0 dB, a higher-quality flex cable is needed as a replacement, which reduces insertion loss.

The differential and single-ended impedance targets are relaxed from the legacy 100 Ω differential and 50 Ω single-ended targets to better reflect design limitations of an ultrathin PCB. This has been verified through simulation.

Table 8-5 MIPI_CSI D-PHY routing constraints (up to 2.5 Gbps)

Metrics		Guidance											
Data rate		500 Mbps/ lane		750 Mbps/ lane		1.0 Gbps/ lane		1.5 Gbps/ lane		2.1 Gbps/ lane		2.5 Gbps/ lane	
Total channel insertion loss		-2.1 dB		-2.3 dB		-2.3 dB		-2.5 dB		-3.5 dB		-6.5 dB	
Ex cable length [a]		3”	6”	3”	6”	3”	6”	3”	6”	3”	6”	3”	6”
Cable insertion loss [b]		-0.5 dB	-1.0 dB	-0.7 dB	-1.15 dB	-0.75 dB	-1.4 dB	-0.9 dB	-1.8 dB	-1.3 dB	-2.3 dB	-2.1 dB	-3.5 dB
Maximum PCB trace length		260 mm	190 mm	210 mm	155 mm	200 mm	125 mm	145 mm	60 mm	170 mm	90 mm	210 mm	150 mm
Impedance	Differential	70 to 100 Ω											
	Single-ended	36 to 52 Ω											

Table 8-5 MIPI_CSI D-PHY routing constraints (up to 2.5 Gbps) (cont.)

Metrics		Guidance	
Length match	Intralane (differential pair)	0.7 mm	
	Interlane (within each MIPI CSI group)	1.4 mm	
Spacing	To all other signals	2.5 × line width	
	Intralane (differential pair)	1 × line width	
	Interlane (within each MIPI CSI group)	1 × line width	1.5 × line width

[a] The flex cable length used in this table is an example with specified insertion loss.
[b] The insertion loss is measured at 0.5 × data rate (Gbps) in GHz. Flex cable insertion loss can be measured using a vector signal analyzer or obtained from the flex-cable datasheet. Cable insertion loss on the design should be no worse than what is listed above.
[c] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.10 MIPI_DSI D-PHY routing constraints (up to 2.5 Gbps)

To interpret this table, use the 500 Mbps/lane as an example. If cable insertion loss is -0.8 dB or better, the trace length needs to be $\leq$ 280 mm. If cable insertion loss is between -0.8 dB and -1.4 dB, the trace length needs to be $\leq$ 210 mm. If cable insertion loss is worse than -1.4 dB, a higher-quality flex cable is needed as a replacement, which reduces insertion loss.

Differential and single-ended impedance targets are relaxed from the legacy 100 Ω differential and 50 Ω single-ended targets to better reflect design limitations of an ultra-thin PCB. This has been verified through simulation.

Table 8-6 MIPI_DSI D-PHY routing constraints (up to 2.5 Gbps)

Metrics		Guidance										
Data rate		500 Mbps/lane		750 Mbps/lane		1.0 Gbps/lane		1.5 Gbps/lane		2.1 Gbps/lane		2.5 Gbps/lane
Total channel insertion loss		-2.1 dB		-2.3 dB		-2.3 dB		-2.5 dB		-3.5 dB		-3.8 dB
Flex cable length [a]		3”	6”	3”	6”	3”	6”	3”	6”	3”	6”	3”
Cable insertion loss [b]		-0.8 dB	-1.4 dB	-1 dB	-1.5 dB	-1.1 dB	-1.7 dB	-1.2 dB	-2.2 dB	-1.6 dB	-2.8 dB	-2.1 dB
Maximum PCB trace length		280 mm	210 mm	210 mm	150 mm	200 mm	100 mm	135 mm	40 mm	150 mm	80 mm	70 mm
Impedance	Differential [c]	70 to 100 Ω										
	Single-ended [c]	36 to 52 Ω										
Length match	Intralane (differential pair)	0.7 mm										
	Interlane (within each MIPI DSI group)	1.4 mm										
Spacing	To all other signal	2 × line width				2.5 × line width						
	Intralane (differential pair)	1 × line width										
	Interlane (within each MIPI DSI group)	1 × line width				1.5 × line width						

[a] Flex cable length used in this table is an example with specified insertion loss.
[b] The insertion loss is measured at 0.5 × data rate (Gbps) in GHz. Flex cable insertion loss can be measured using a vector signal analyzer or obtained from the flex-cable datasheet. Cable insertion loss on the design should be no worse than what is listed above.
[c] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.11 C-PHY (DSI and CSI) routing constraints

To interpret the following table, use the 1.5 Gsps/lane as an example. If the cable insertion loss is -0.9 dB or better, the trace length needs to be $\leq$ 250 mm. If cable insertion loss is between -0.9 dB and -1.8 dB, the trace length needs to be $\leq$ 165 mm. If cable insertion loss is worse than -1.8 dB, a higher-quality flex cable is needed as a replacement, which reduces insertion loss.

The single-ended impedance targets are relaxed from the legacy 50 Ω single-ended targets to better reflect design limitations of an ultra-thin PCB. This has been verified by simulation.

Table 8-7 C-PHY (DSI and CSI) routing guidelines

Metrics		Guidance														
Data rate		1.5 Gsps/lane			2.1 Gsps/lane			2.5 Gsps/lane			3.5 Gsps/lane			4.5 Gsps/lane		
Flex cable length (in inches) [a]		1"	3"	6"	1"	3"	6"	1"	3"	6"	1"	3"	6"	1"	3"	6"
Flex cable insertion loss [b]		-0.4 dB	-0.9 dB	-1.8 dB	-0.5 dB	-1.3 dB	-2.3 dB	-0.48 dB	-1.4 dB	-2.5 dB	-0.71 dB	-1,46 dB	-3.12 dB	-0.48 dB	-1.72 dB	-3.8 dB
Maximum PCB trace length		280 mm	220 mm	140 mm	225 mm	150 mm	85 mm	200 mm	130 mm	40 mm	208 mm	188 mm	152 mm	168 mm	132 mm	88 mm
Maximum PCB IL – Estimated		-2.5 dB	-2.0 dB	-1.3 dB	-2.5 dB	-1.7 dB	-1.0 dB	-2.2 dB	-1.4 dB	-0.5 dB	-3.69 dB	-3.34 dB	-2.08 dB	-3.46 dB	-2.88 dB	-1.1 dB
Impedance	Single-ended [c]	36 to 52 Ω														
Length match	Intralane (within a trio)	0.7 mm														
	Interlane (trio-to-trio)	4.2 mm														
Spacing	To all other signals	3 × line width														
	Intralane (within a trio)	1.5 × line width														
	Interlane (trio to trio)	1.5 × line width														

[a] The flex cable length used in this table is an example with specified insertion loss.
[b] The insertion loss is measured at 0.5 × data rate (in Gsps) in GHz. Flex cable insertion loss can be measured using a vector signal analyzer or obtained from the flex-cable datasheet. Cable insertion loss on the design should be no worse than what is listed above.
[c] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.12 PCIe layout guidelines

The following are the general PCIe routing guidelines:

- All other sensitive/high-speed signals and circuits must be protected from PCIe corruption.
- PCIe signals must be protected from noisy signals (clocks, SMPS, and so forth).
- Pay extra attention to crosstalk, ISI, and intralane skew and impedance discontinuities.
- For a single-board design, PCIe Tx AC coupling capacitors can be anywhere along the line, but better be placed close to source or receiver side to keep good SI of main route on PCB.
- Each trace needs to be adjacent to a ground plane.
- To maintain impedance balance, maintain positive and negative traces as balanced as possible in terms of the signal and its return path.
- Avoid broadside coupling and routing a signal trace directly over another signal trace.
- To reduce the probability for layer-to-layer manufacturing variation, minimize layer transitions on the main route (in other words, apply layer transitions only at QRB5165 device breakouts and connectors to ensure minimum layer transitions on the main route).

Additional layout guidelines

- The trace length matching between the reference clock, Tx, and Rx pairs are not required. The trace serpentines are not required.
- Ensure not to stagger the capacitors. This can affect the differential integrity of the design and can create EMI.
- In the case of serpentines, one line of a differential pair must be routed to make up a length delta, then it must be routed at the source (breakout) – this ensures that lines stay differential thereafter.

The following table lists the PCIe routing constraints.

Table 8-8 PCIe routing constraints

Metrics			Guidance	
Data rate			8 Gbps (Gen 3.0)	5 Gbps (Gen 2.0)
Channel differential insertion loss			-10 dB at 4 GHz	-6 dB at 2.5 GHz
Bus length			300 mm	250 mm
Impedance [a]	Differential	Field route	70 to 100 Ω	
Length match	Intralane length match		5 ps (0.7 mm)	
Spacing	To all other signals	Field route	4 × line width	
	Rx-to-Tx		4 × line width	3 × line width
Component	AC capacitor		176 to 265 nF	75 to 265 nF

[a] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.13 SDC2 external SD card routing constraints

The following table lists the SDC2 external SD card routing constraints. Note that the guidance of bus length is based on the SD card I/O driver capability and maximum load capacitance per SD card specifications. The QRB5165 SDIO pad capability can drive longer bus if needed.

Table 8-9 SDC2 external SD card routing constraints

Metrics			Guidance		Comments
Clock frequency			50 MHz DDR/100 MHz SDR	208 MHz	
Data rate			50 Mbps	104 Mbps	
SD card capacitive load			10 pF	5 pF [a]	For UHS104 modes, if card capacitance is > 5 pF then trace length must be reduced accordingly.
Bus length			< 150 mm	< 50 mm	
Impedance [a]		Field route	36 to 50 Ω		
Clock to data length matching			6 mm	2 mm	
Spacing	To other signals	Field route	1.5 × line width		
	Lane-to-lane		1.5 × line width		
Component	CLK series termination resistor		18 – 27 Ω		

[a] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.14 UFS routing constraints

The following table lists the UFS routing constraints. Note that there is no Rx-to-Tx interlane matching requirement.

Table 8-10 UFS routing constraints

Metrics			Guidance			
Data rate			6.0 Gbps (Gear 3)			10 Gbps (Gear 4, Rate A)
Channel differential insertion loss			-6 dB at 3 GHz			-1.3 dB at 2.9 GHz, -5.0 dB at 5.8 GHz
Bus length			25 mm	75 mm	150 mm	≤ 70 mm
Impedance [a]	Differential	Field route	75 to105 Ω			70 to 110 Ω
Length match	Intralane length match		5 ps (0.7 mm)			
	Tx 1 to Tx 2 or Rx 1 to Rx 2		30 ps (4.2 mm)			
Spacing	Rx-to-Tx	Field route	1.5 × line width	2 × line width	3 × line width	2.5 × line width

Table 8-10 UFS routing constraints (cont.)

Metrics			Guidance			
	Rx1-to-Rx2 or Tx1-to-Tx2		NA	NA	NA	2 × line width
	To all other signals		2 × line width	2 × line width	3 × line width	3 × line width

[a] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

8.2.15 SPMI layout rules

The following table lists the SPMI routing constraints.

Table 8-11 SPMI layout rules

Metrics			Guidance			
Date rate			19.2 Mbps			
Bus length			< 60 mm	< 90 mm	< 120 mm	< 150 mm
Routing topology			Star routing possible . Daisy chain preferred.		Daisy chain	
Stub length			<5 mm	< 3 mm	< 3 mm	< 3 mm
Loading	Bus capacitance		Min 5 pF to max 20 pF lumped capacitance			
Impedance [a]	SE	Field route	45 – 55 Ω	36 – 44 Ω	36 – 44 Ω	36 – 44 Ω
Length match	Clk to data		<2 mm (15 pS)			
Spacing	Clk to data	Field route	1.5 × line width		2 × line width	
	To other signals					

[a] The range of PCB impedance includes manufacturing variation. The PCB designer is responsible for working with their PCB vendor to ensure each design falls within the published impedance range.

9 Reference documents

The following table lists the documents which cover the technical information for QRB5165 device. The released QRB5165 documents are posted on the Qualcomm.com website and are available for download.

Table 9-1 QRB5165 reference documents

Document number	Document title
80-PV086-1	*QRB5165 Data Sheet*
80-PV086-1A	*QRB5165 Pin Assignment and GPIO Configuration Spreadsheet*
80-PV086-4	*QRB5165 Device Revision Guide*
80-PV086-5P (this document)	*QRB5165 Hardware Design Guide*
80-PV086-2X	*QRB5165 Hardware Register Description*
80-PV086-97	*QRB5165 QFPROM Programming Reference Guide*
80-PV086-111	*QRB5165 Schematic Review Checklist*
80-PV086-112	*QRB5165 Layout Review Checklist*
80-VP300-16	*PoP Memory SM8250/SXR2130/QRB5165/QCS8250/QCS8245 Recommendations*
80-VT310-13	*Power Delivery Network Design Training*
80-VU794-14	*Coefficient of Thermal Spreading (CTS) – Figure of Merit for Mobile Thermal Management*
80-VU794-16	*Mobile Devices Hardware Thermal Management*
80-VU794-24	*Design for Thermal: Key Requirements Why, What, Where, When, and How*
80-VL764-131	*QRD8250 Bill of Materials (BOM)*
80-VL771-131	*QRD8250_EU Bill of Materials (BOM)*
DP10-VL764-100A	*Design Package, QRD8250, Smart Phone, Electrical Hardware, PCB Schematic and Layout*
DP10-VL771-100A	*Design Package, QRD8250_EU, Smart Phone, Electrical Hardware, PCB Schematic and Layout*
80-PV086-41	*QRB5165 RB5 SoM Design Example Schematic*
DP25-PV086-42	*Design Package, QRB5165 RB5 Main Board*
DP25-PV086-43	*Design Package, QRB5165 RB5 Navigation Mezzanine*
DP25-PV086-45	*Design Package, QRB5165 RB5 5G Mezzanine*

Table 9-1 QRB5165 reference documents (cont.)

Document number	Document title
80-PV086-300	*QRB5165 Linux Ubuntu Quick Start Guide*
80-PV086-600	*QRB5165 RB5 Hardware Reference Platform User Guide*

10 Revision history

The following table lists the technical content changes for all revisions.

Revision	Date	Description
AA	July 2023	Initial release

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